



Uma introdução à fenotipagem de imagens

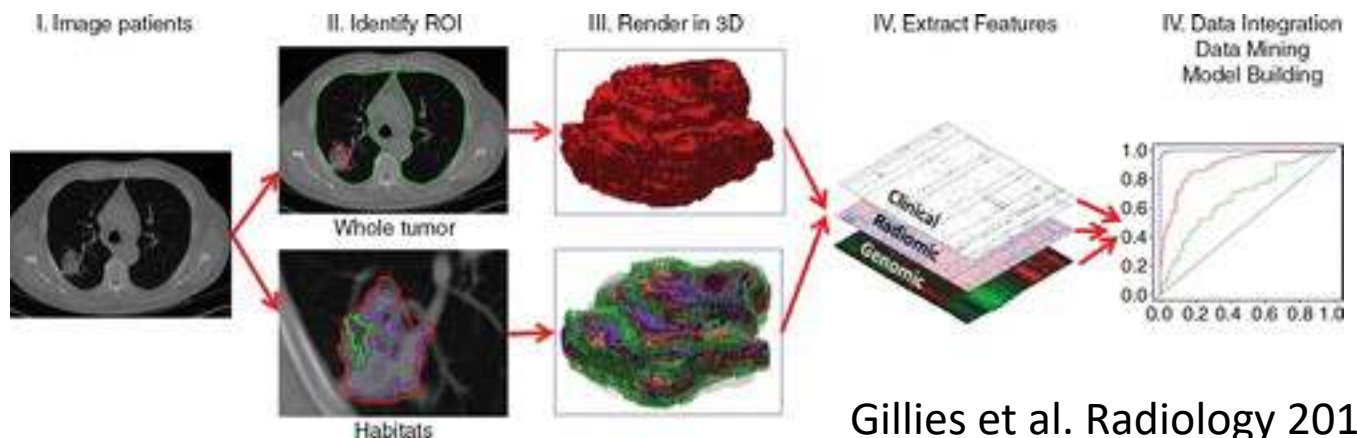
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Professora Assistente de Radiologia
Diretora de Treinamento e Disseminação
Neuroimaging Analysis Center

Brigham and Women's Hospital
Harvard Medical School

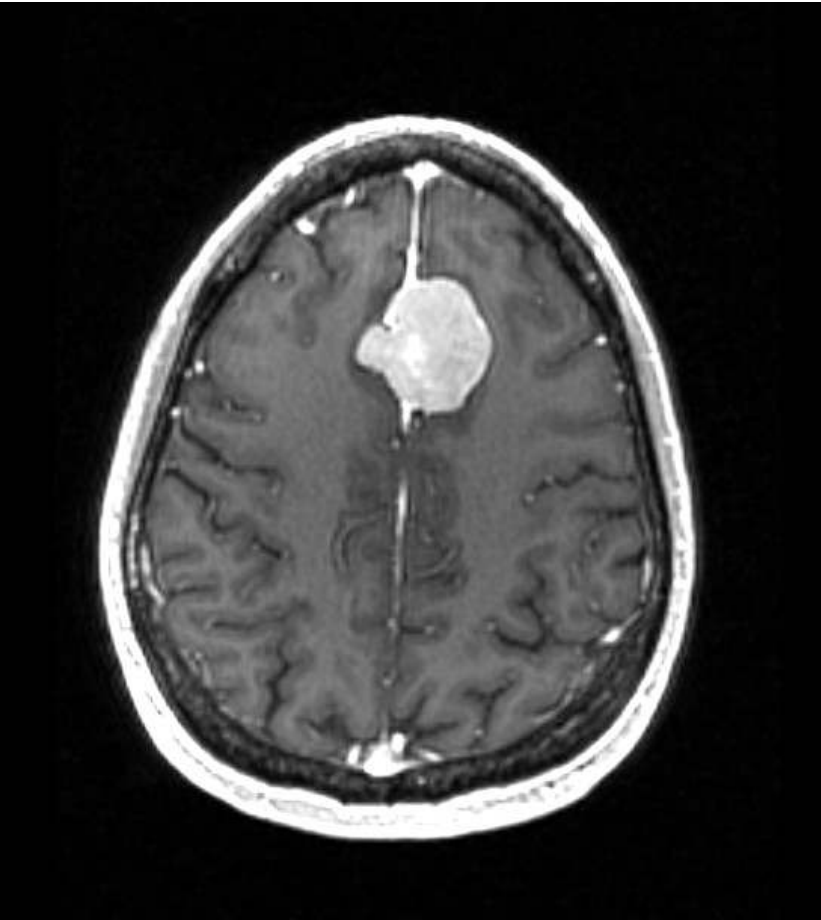
Fenótipos de imagem

- Os fenótipos de imagem descrevem características da doença que podem ser detectadas por meio de imagens médicas combinadas com detecção de características, aprendizado de máquina e análise estatística, e correlacionadas com outros indicadores da doença.



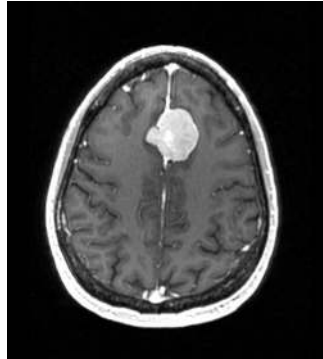
Gillies et al. Radiology 2015

Caso clínico: meningioma

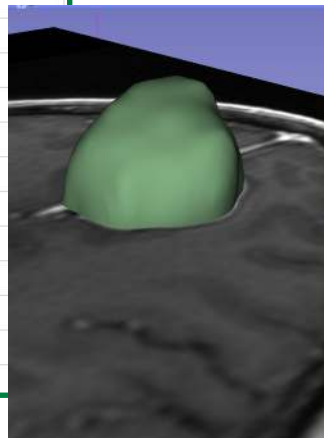


- Os meningiomas são tumores cerebrais extra-axiais de crescimento lento que surgem das células aracnoides
- As opções de tratamento incluem observação, cirurgia e radioterapia
- Preditores de graus tumorais baseados em imagens têm o potencial de aprimorar a tomada de decisão clínica

Objetivo geral

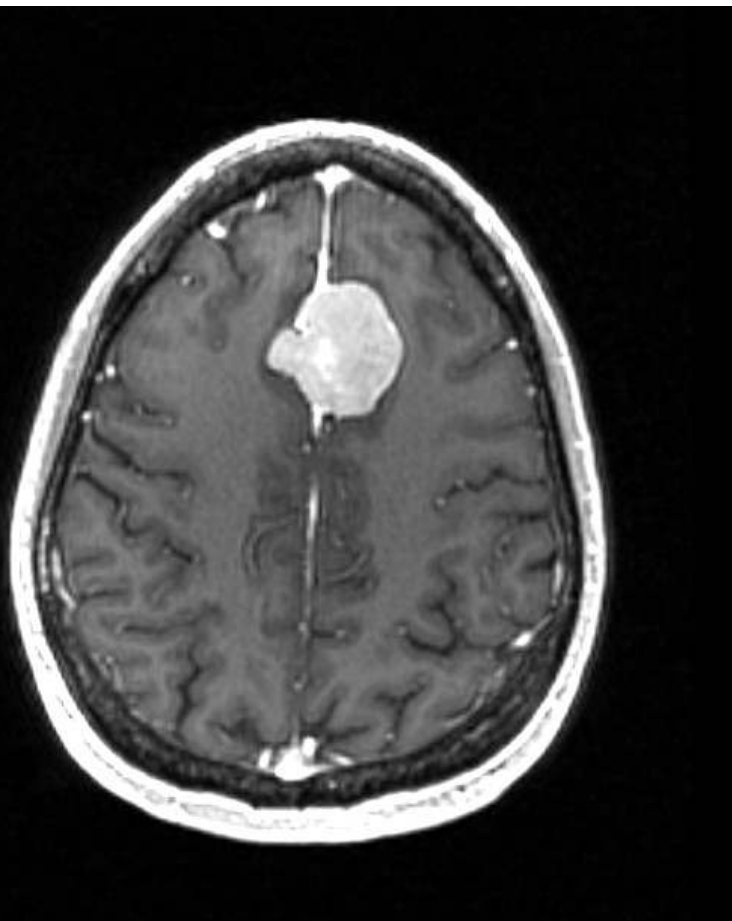


Kurtosis	12.4315749384
Mean	179.297568367
GrayLevelVariance	1.00567651545
HighGrayLevelEmphasis	45.873151021
DependenceEntropy	6.16459662954
DependenceNonUniformity	548.562883325
GrayLevelNonUniformity	4972.61959737
SmallDependenceEmphasis	0.0389261318031
SmallDependenceHighGrayLe...	1.92568883112
DependenceNonUniformityNor...	0.0395817074338
LargeDependenceEmphasis	237.650263367
LargeDependenceLowGrayLev...	5.58612453125
DependenceVariance	51.1747739691



Este tutorial é uma introdução aos fundamentos da fenotipagem de imagens para a caracterização de tumores usando a plataforma 3D Slicer.

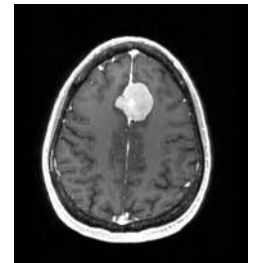
Características de imagem



- As características quantitativas derivadas de dados de imagem têm o potencial de fornecer informações clinicamente relevantes para prever o grau do tumor e avaliar a resposta ao tratamento

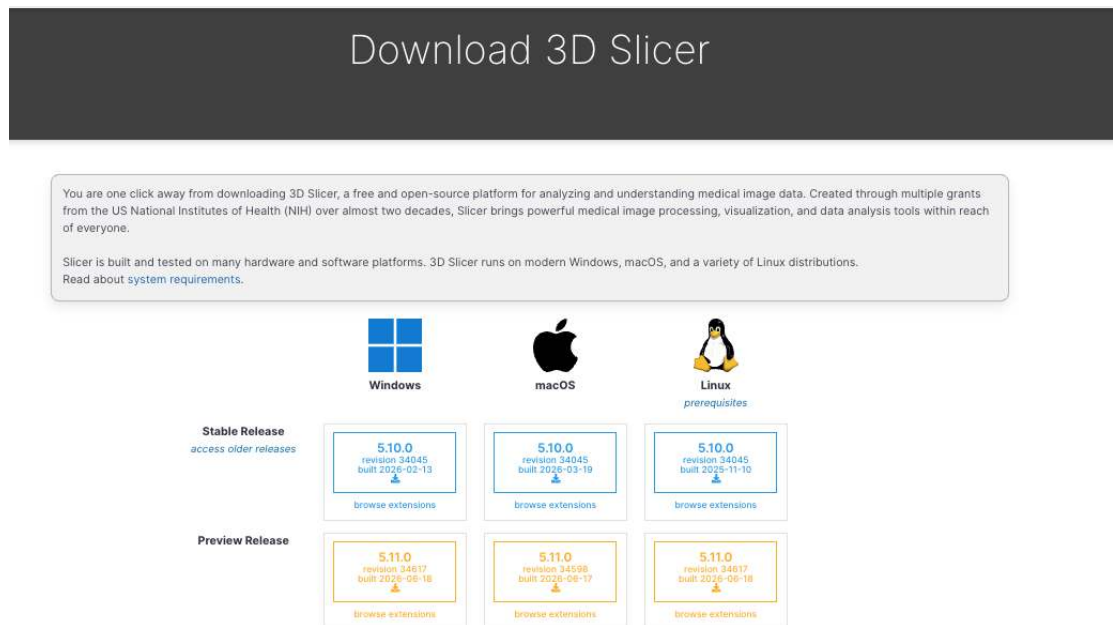
Materiais do tutorial

- 3D Slicer versão 4.10.1
- Slicer Radiomics Extension
- Conjunto de dados Meningioma



Instalação do 3D Slicer

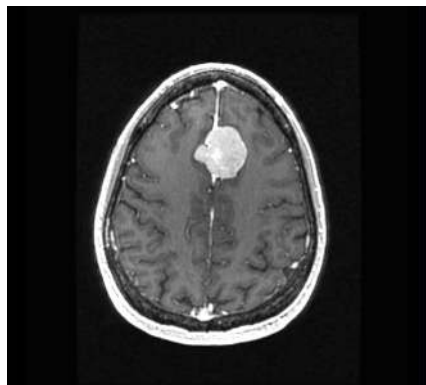
- Para instalar e iniciar o software 3D Slicer no seu computador, siga as instruções do **tutorial Quick Start Guide**, disponível em <https://download.slicer.org/>



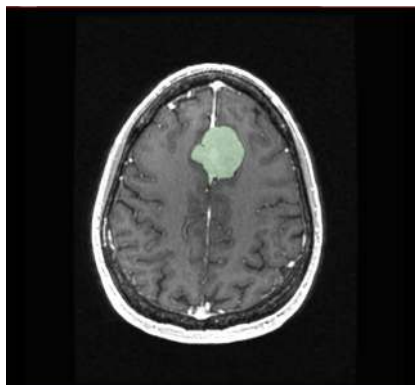
Aviso

- O 3D Slicer é um software livre e de código aberto para pesquisa em computação de imagens médicas, distribuído sob uma licença de estilo BSD.
- O software não é aprovado pela FDA nem possui marcação CE, e destina-se apenas a uso em pesquisa.

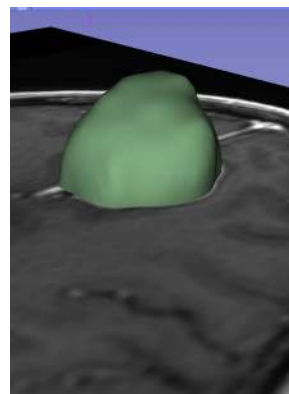
Visão geral do fluxo de trabalho



Etapa 1: Carregamento dos dados



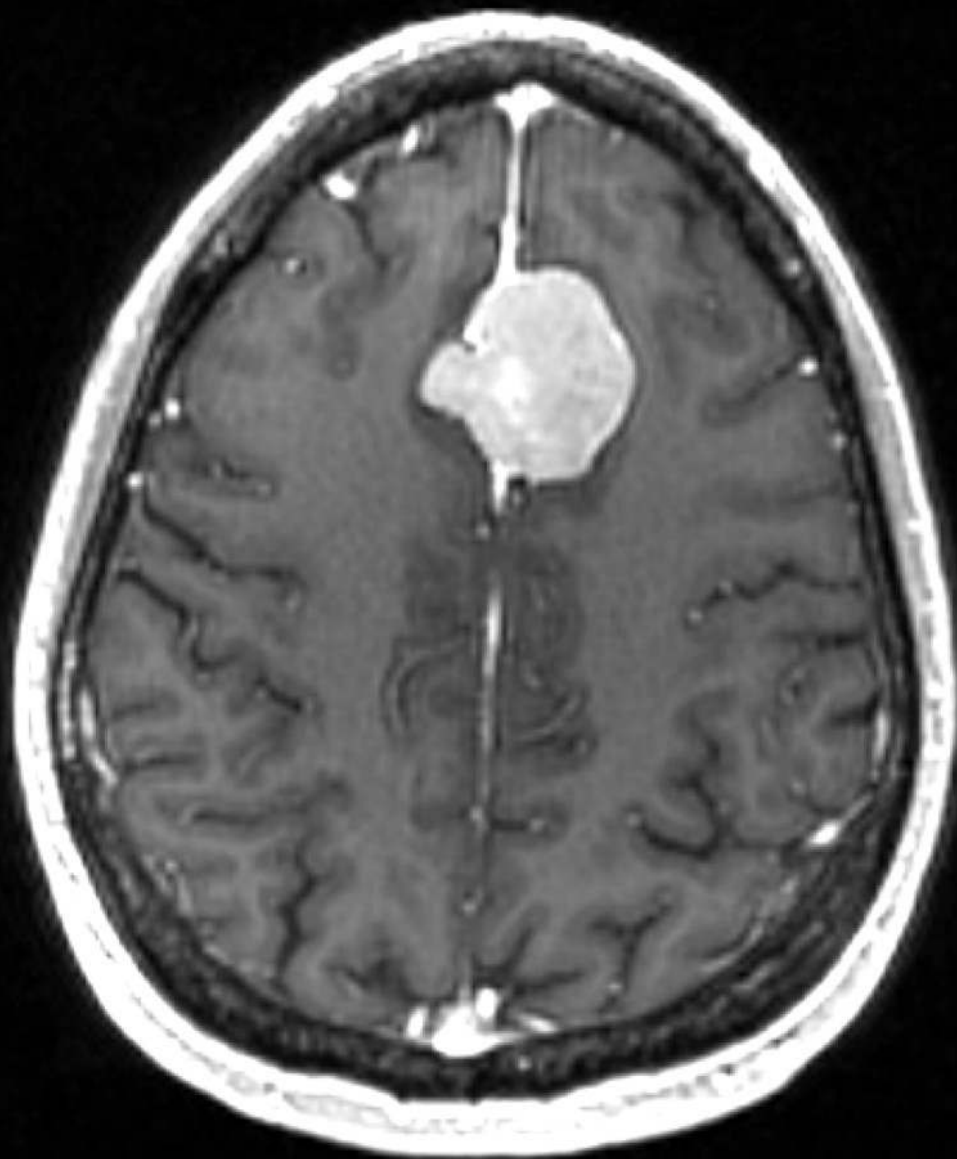
Etapa 2: Segmentação do tumor



Etapa 3: Cálculo do volume do tumor

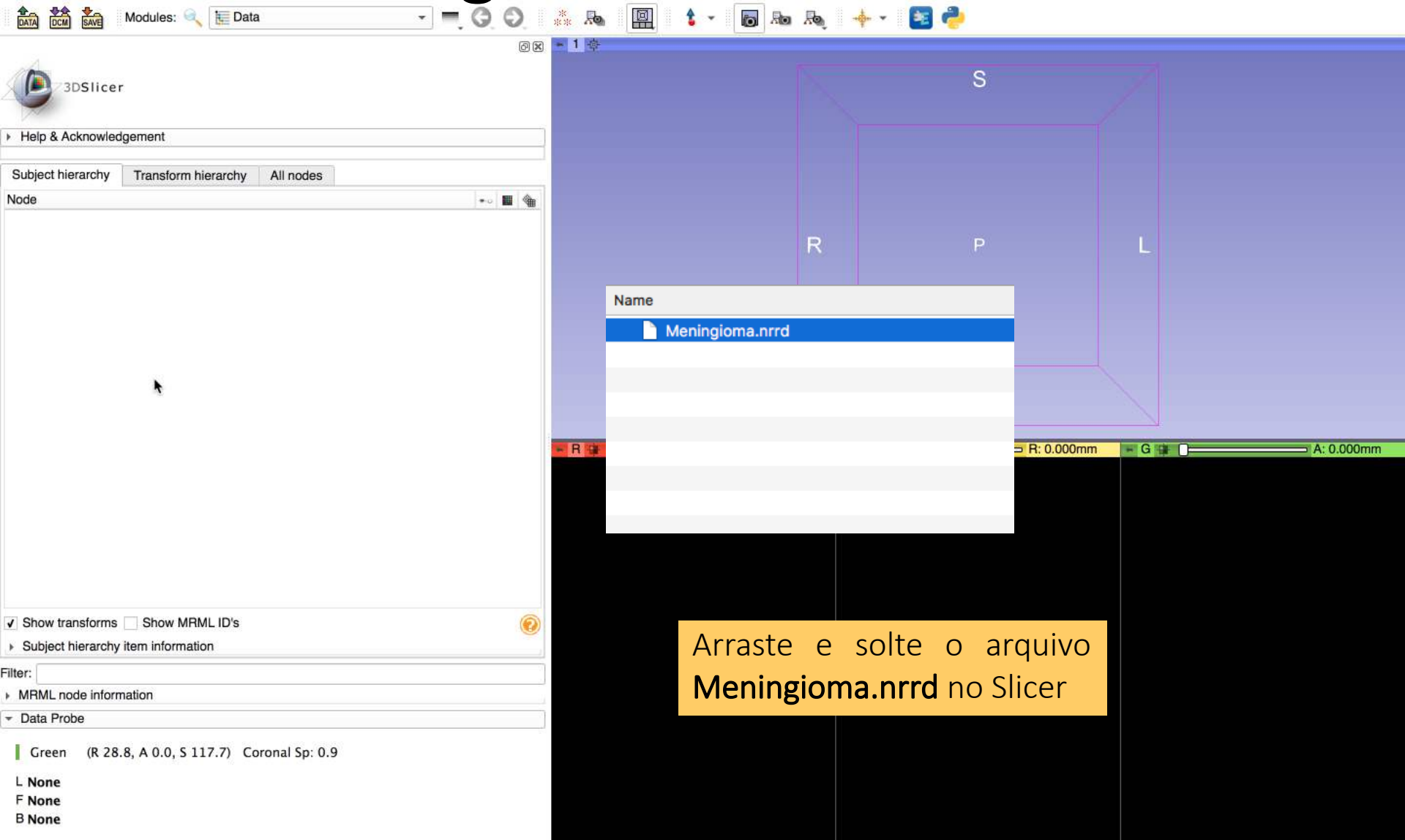
Kurtosis	12.4315749384
Mean	179.297568367
GrayLevelVariance	1.00567651545
HighGrayLevelEmphasis	45.873151021
DependenceEntropy	6.16459662954
DependenceNonUniformity	548.562883325
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SmallDependenceEmphasis	0.0389261318031
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DependenceNonUniformityNor...	0.0395817074338
LargeDependenceEmphasis	237.650263367
LargeDependenceLowGrayLev...	5.58612453125
DependenceVariance	51.1747739691

Etapa 4: Extração das características de imagem

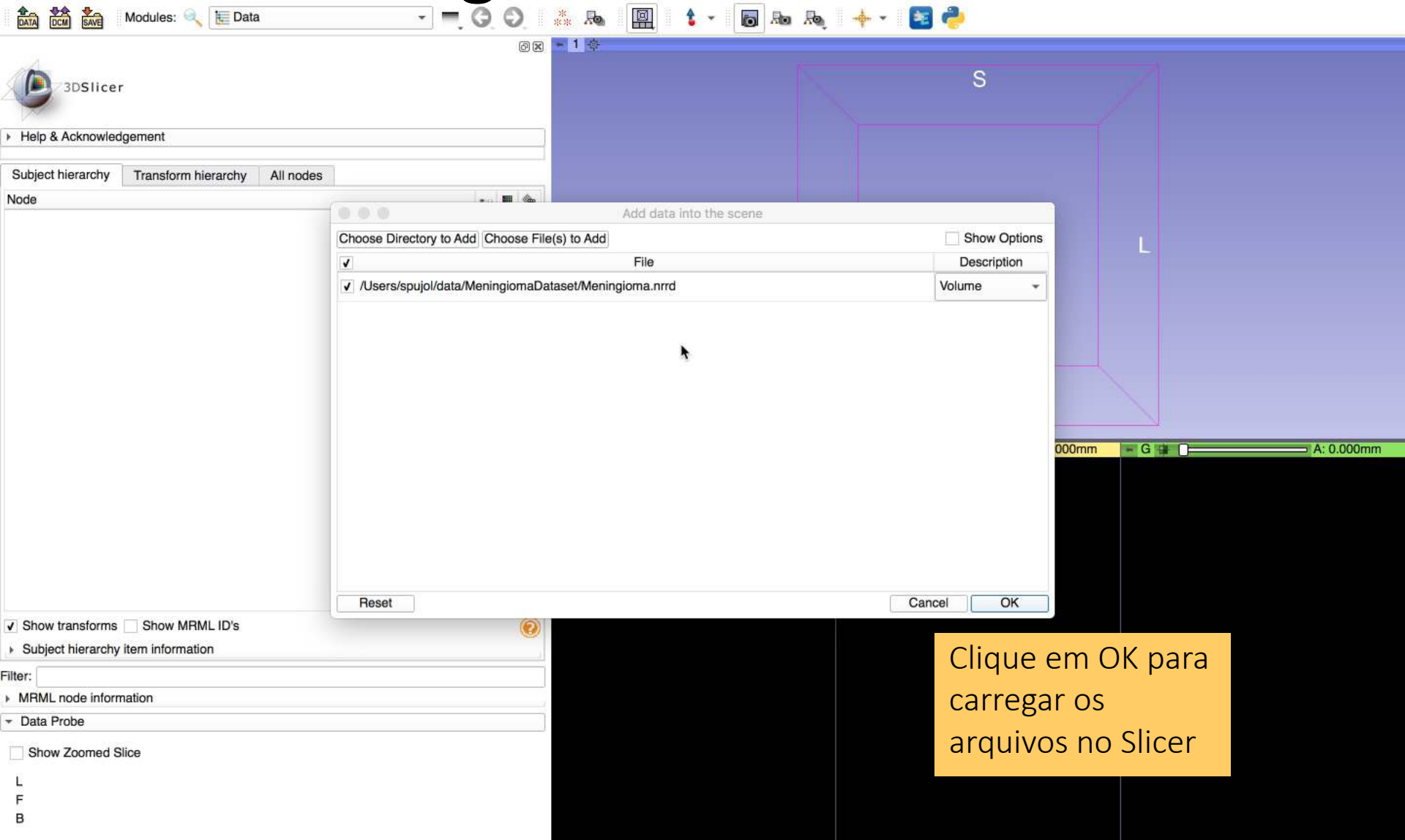


Parte 1:
Carregamento dos
dados e medições
do diâmetro do
tumor

Carregamento dos dados



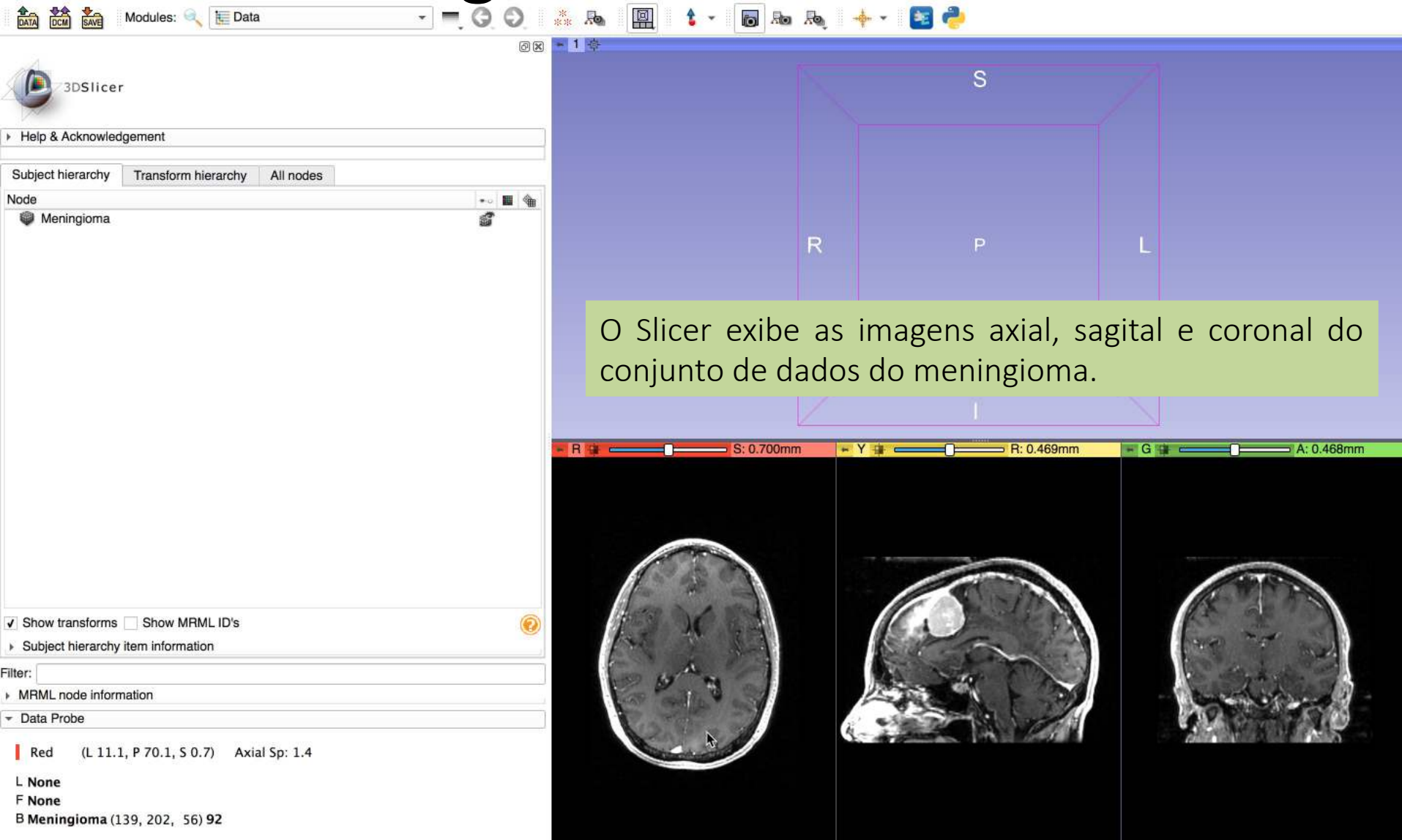
Carregamento dos dados



Clique em OK para
carregar os
arquivos no Slicer



Carregamento dos dados

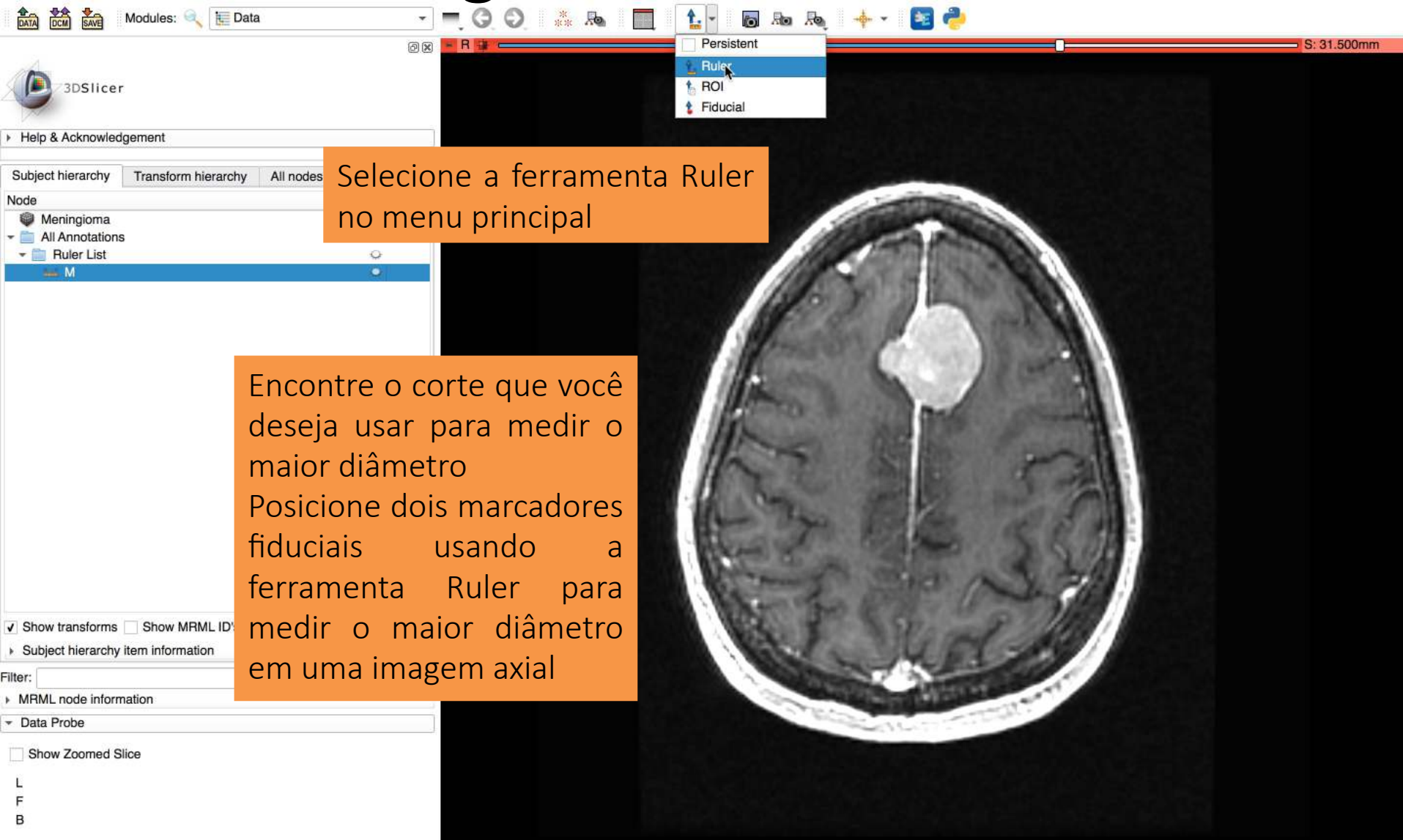


Carregamento dos dados

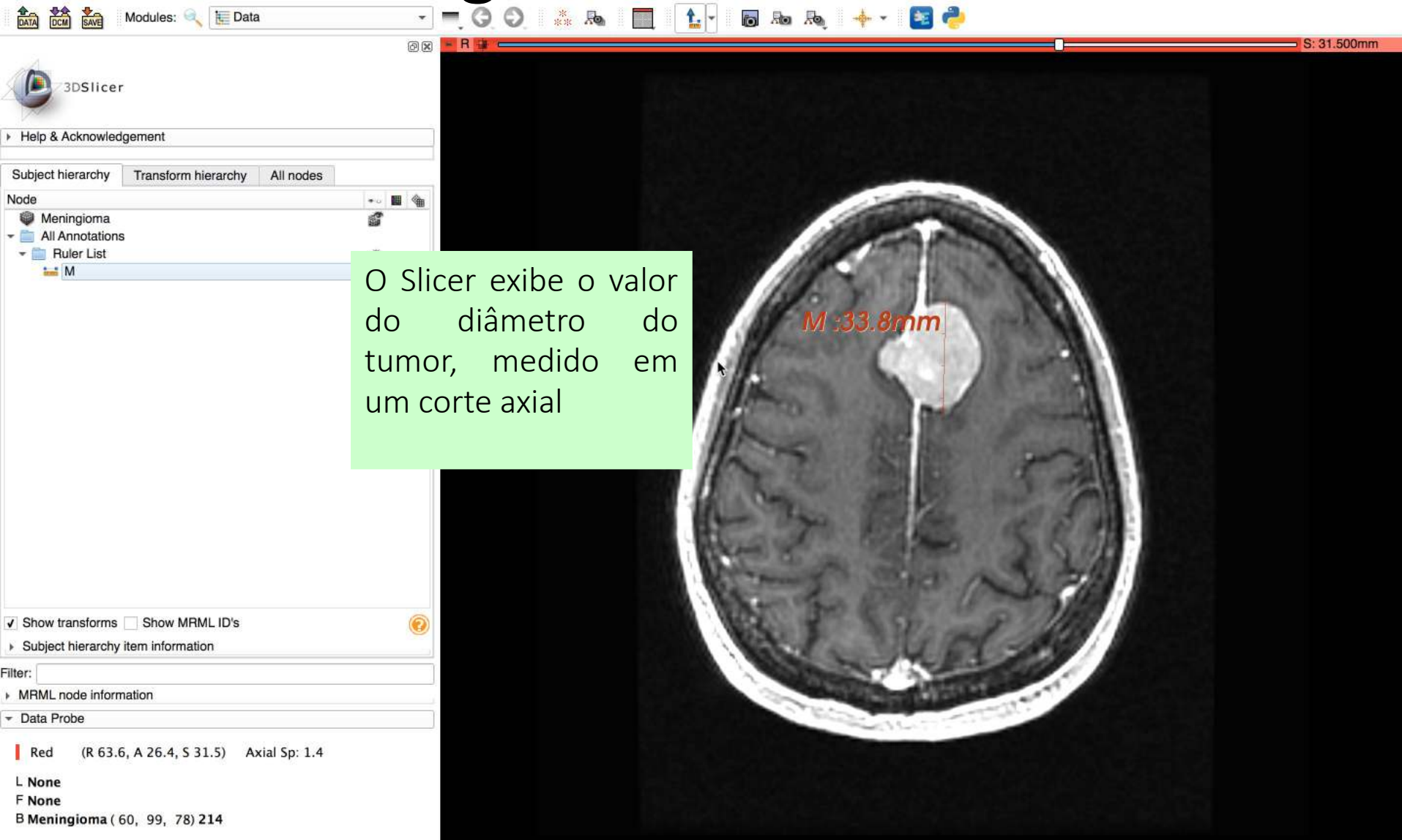
The screenshot displays the 3D Slicer software interface. On the left, the 'Subject hierarchy' panel shows a node named 'Meningioma'. The 'Modules' dropdown menu is open, showing a list of modules including 'Conventional', 'Conventional Widescreen', 'Conventional Quantitative', 'Four-Up', 'Four-Up Table', 'Four-Up Quantitative', 'Dual 3D', 'Triple 3D', '3D only', '3D Table', 'One-Up Quantitative', 'Red slice only', 'Yellow slice only', 'Green slice only', 'Tabbed 3D', 'Tabbed slice', 'Compare', 'Compare Widescreen', 'Compare Grid', 'Three over three', 'Three over three Quantitative', 'Four over four', 'Two over two', 'Side by side', 'Four by three slice', 'Four by two slice', and 'Three by three slice'. The 'Red slice only' option is highlighted. On the right, a 3D brain model is shown with a purple wireframe box indicating a region of interest. Below the 3D model, three 2D slice views (axial, sagittal, and coronal) are displayed. The status bar at the bottom shows coordinates: 'Red (L 11.1, P 70.1, S 0.7)', 'L None', 'F None', and 'B Meningioma (139, 202, 56)'. An orange text box at the bottom center contains the text: 'Selecione o layout Red Slice Only no menu Layout'.

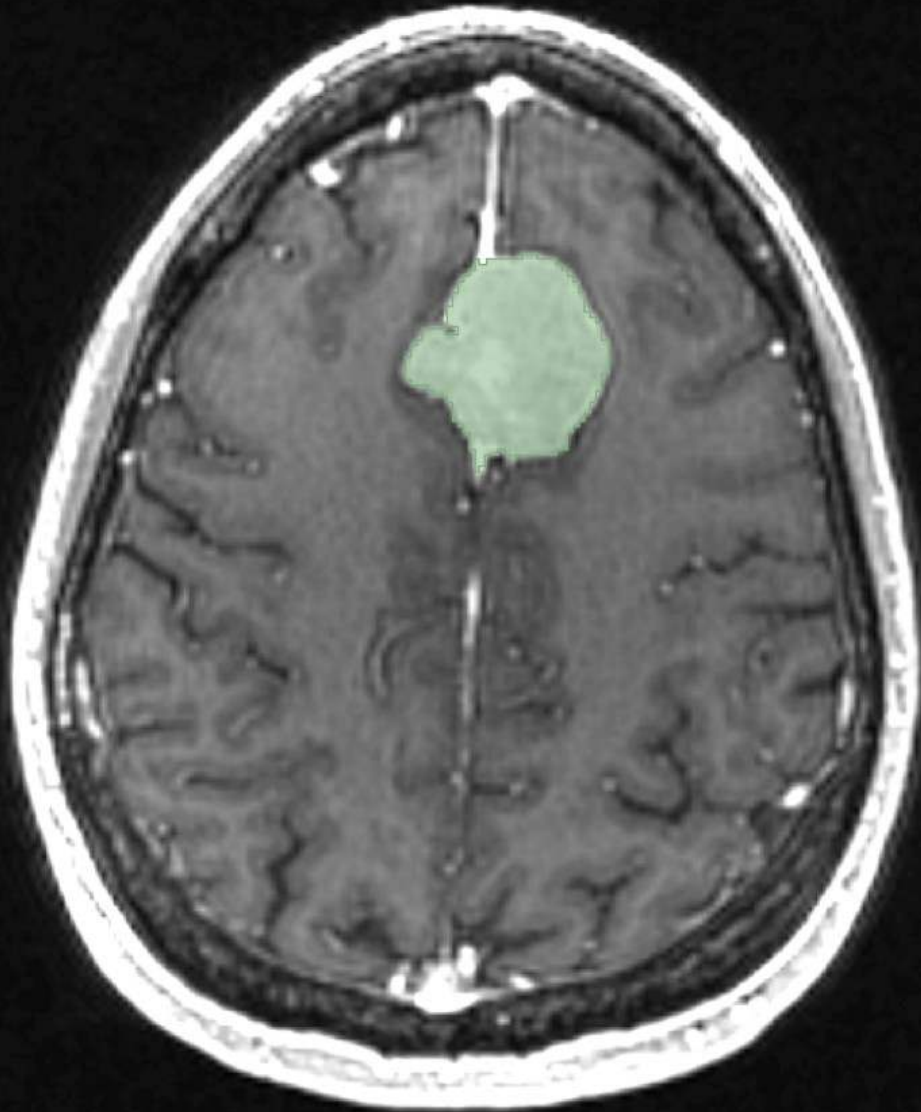
Selecione o layout Red Slice Only no menu Layout

Carregamento dos dados



Carregamento dos dados





Par
Seg

Parte 2: Segmentação do tumor

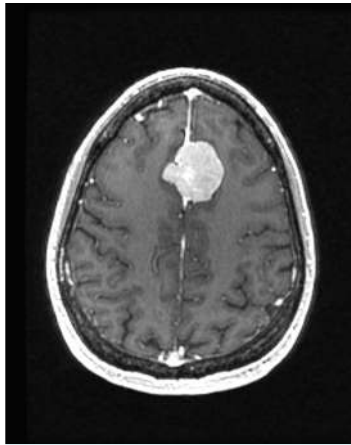
Segmentação de imagens

- A segmentação de imagens é o processo de identificar os contornos de estruturas de interesse nos dados de imagem
- A segmentação de imagens pode ser realizada por contorno manual ou usando algoritmos de segmentação automatizados

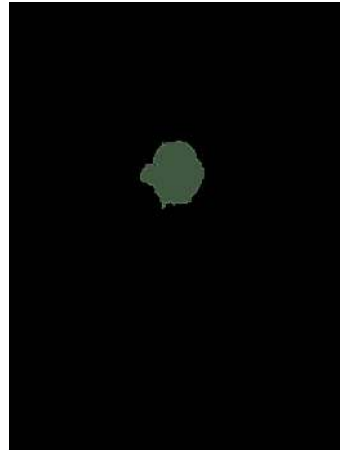
Segmentação de imagens

- O módulo Segment Editor do 3D Slicer oferece ferramentas poderosas para segmentação manual e semiautomática
- O módulo recebe uma imagem de referência (**Master Volume**) como entrada e produz uma imagem segmentada (**Segmentation**) como saída

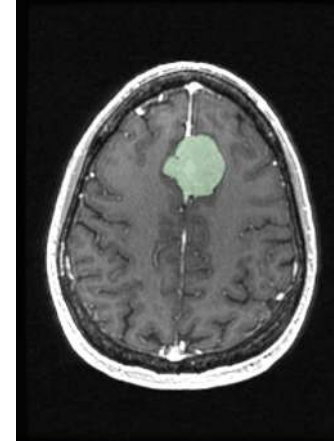
Terminologia



Master Volume



Segmentation

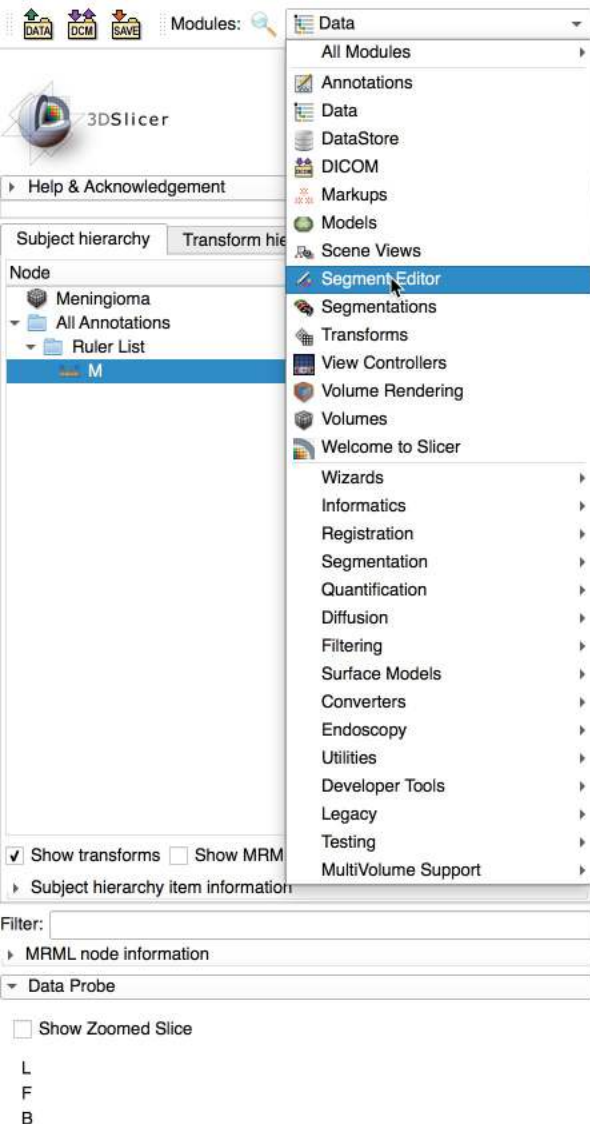


Master Volume
&
Segmentation

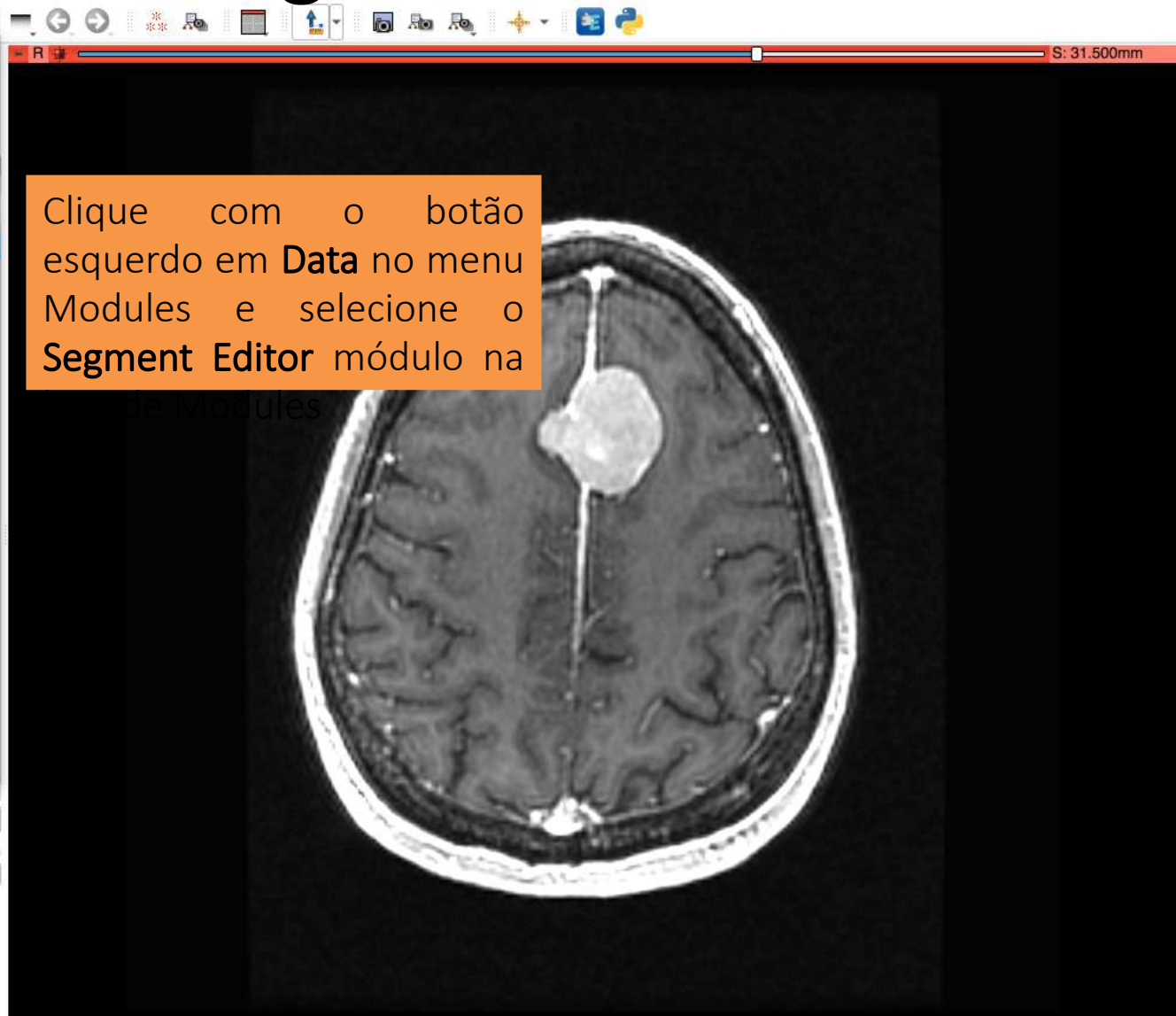
Neste tutorial,

- O Master Volume **Meningioma** é o conjunto de dados inicial de RM cerebral
- O volume Segmentation **Segmentation** é um objeto de segmentação com as mesmas dimensões e origem que o Master Volume

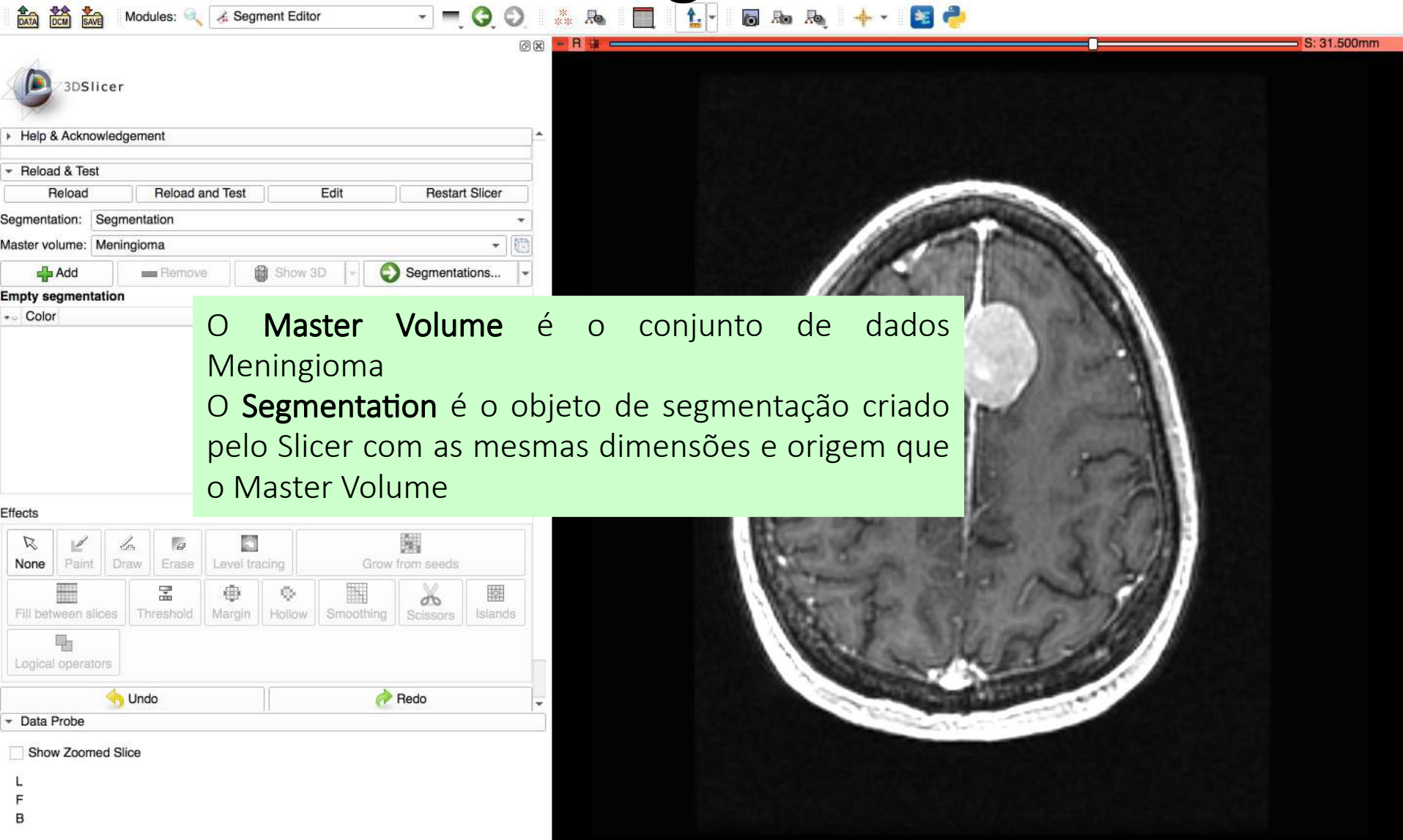
Módulo Segment Editor



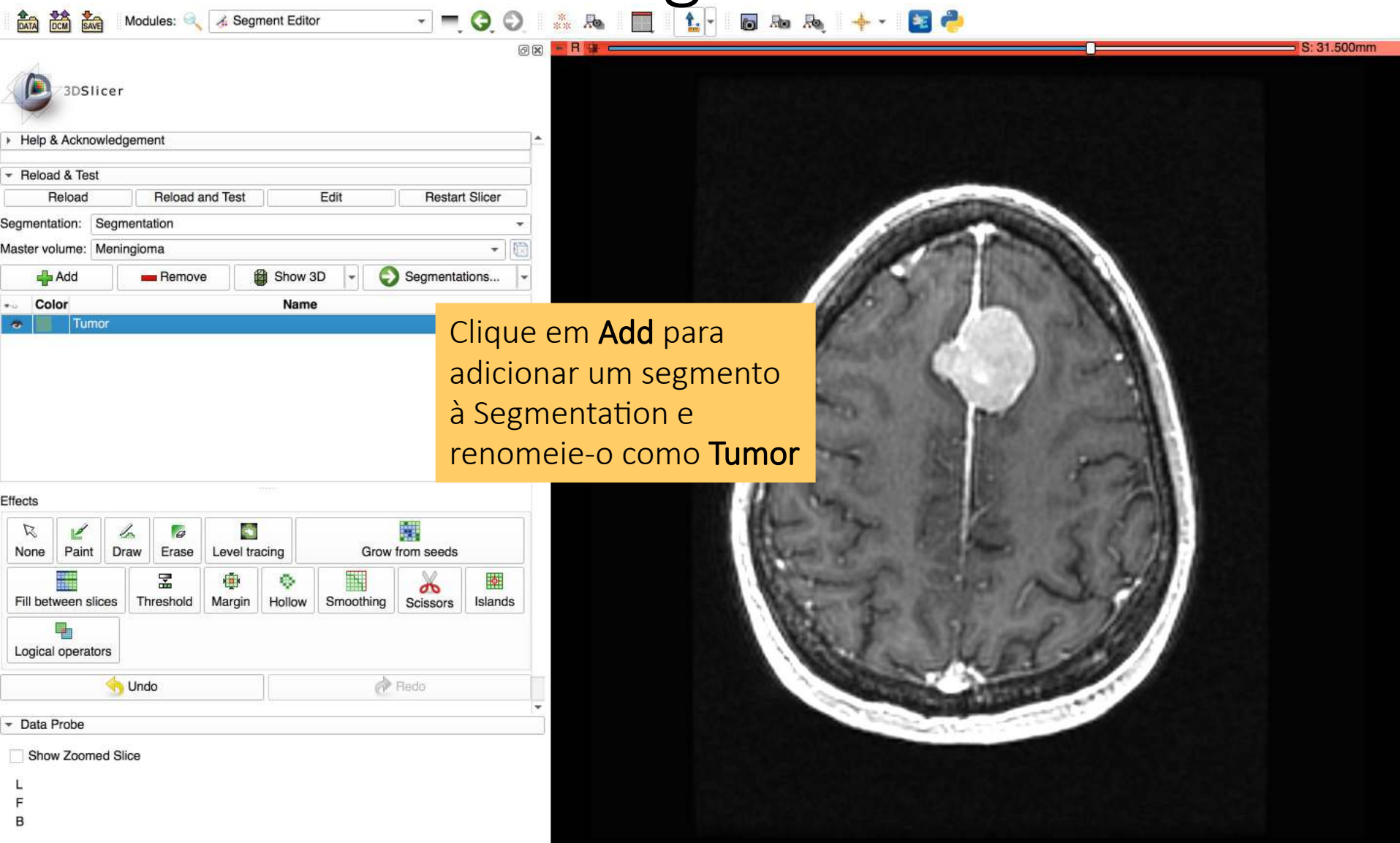
Clique com o botão esquerdo em **Data** no menu Modules e selecione o **Segment Editor** módulo na



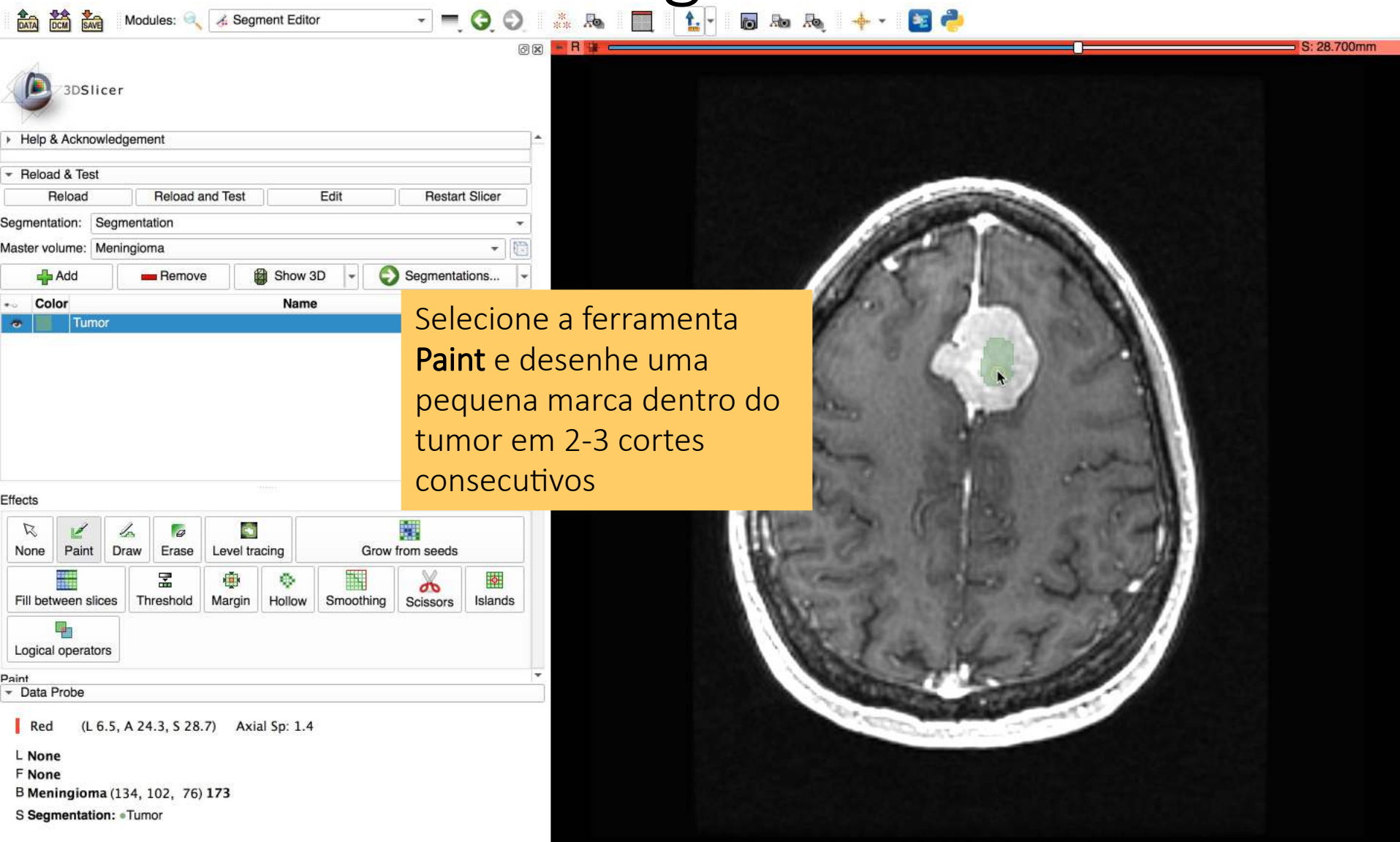
Módulo Segment Editor



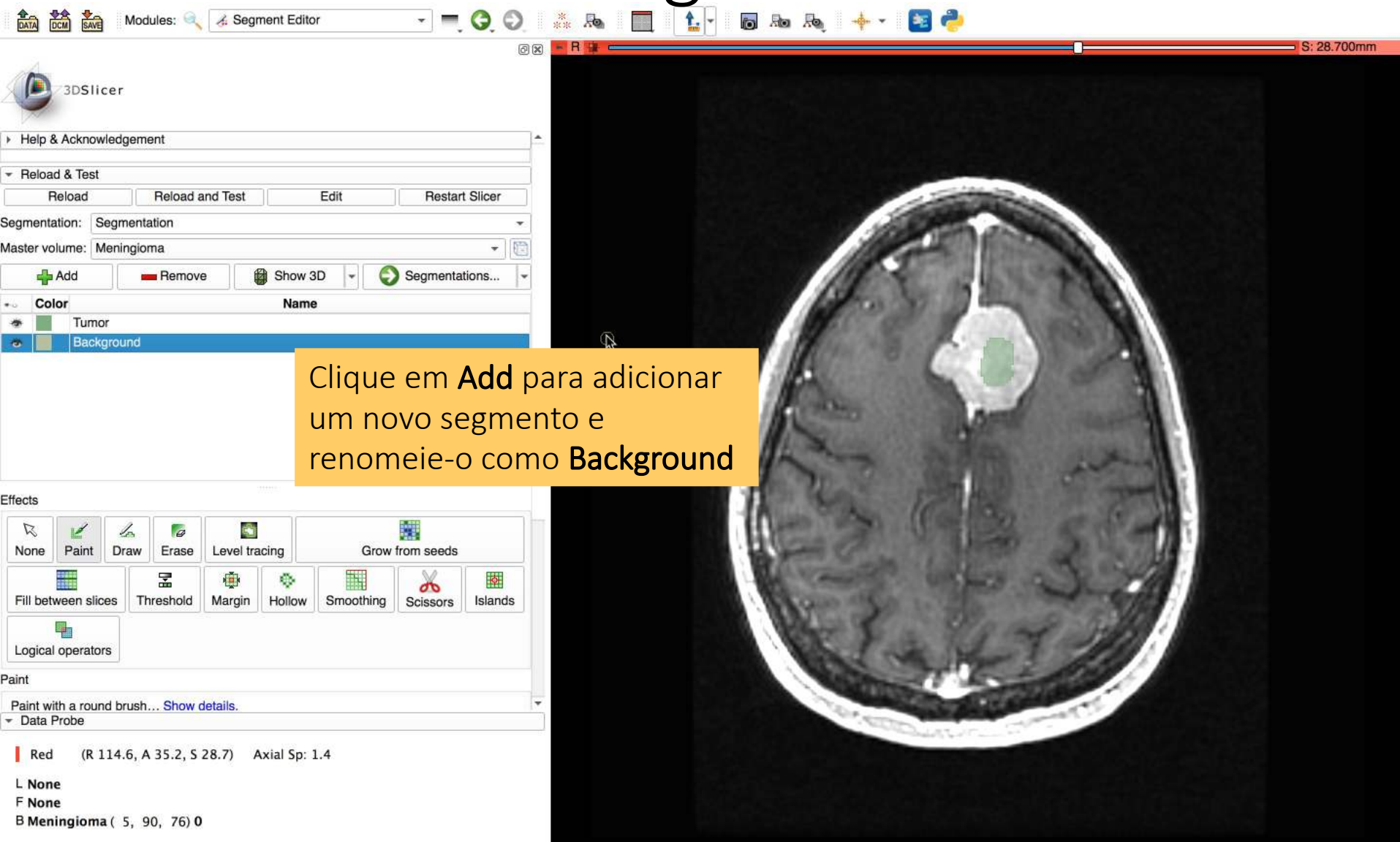
Módulo Segment Editor



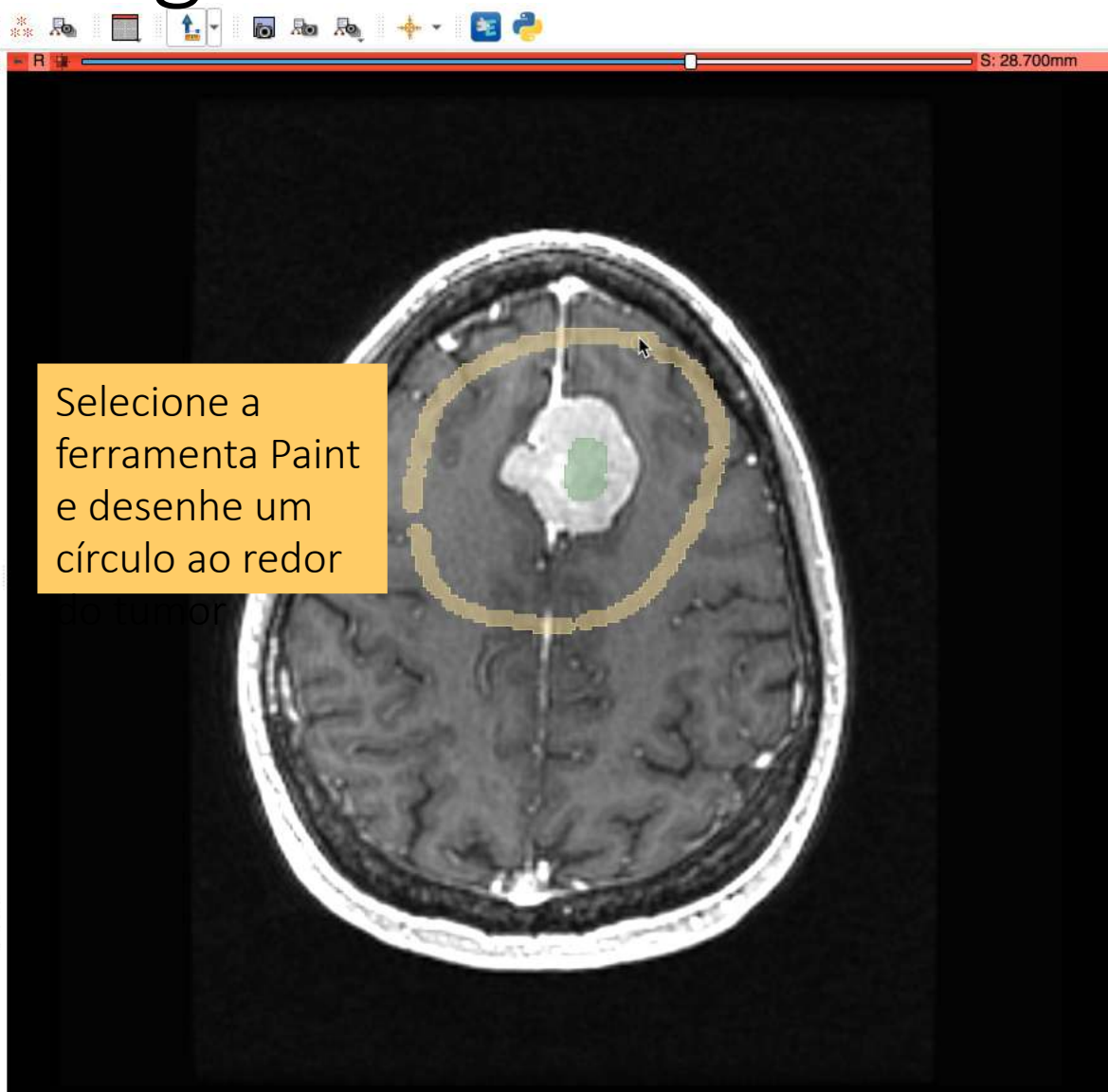
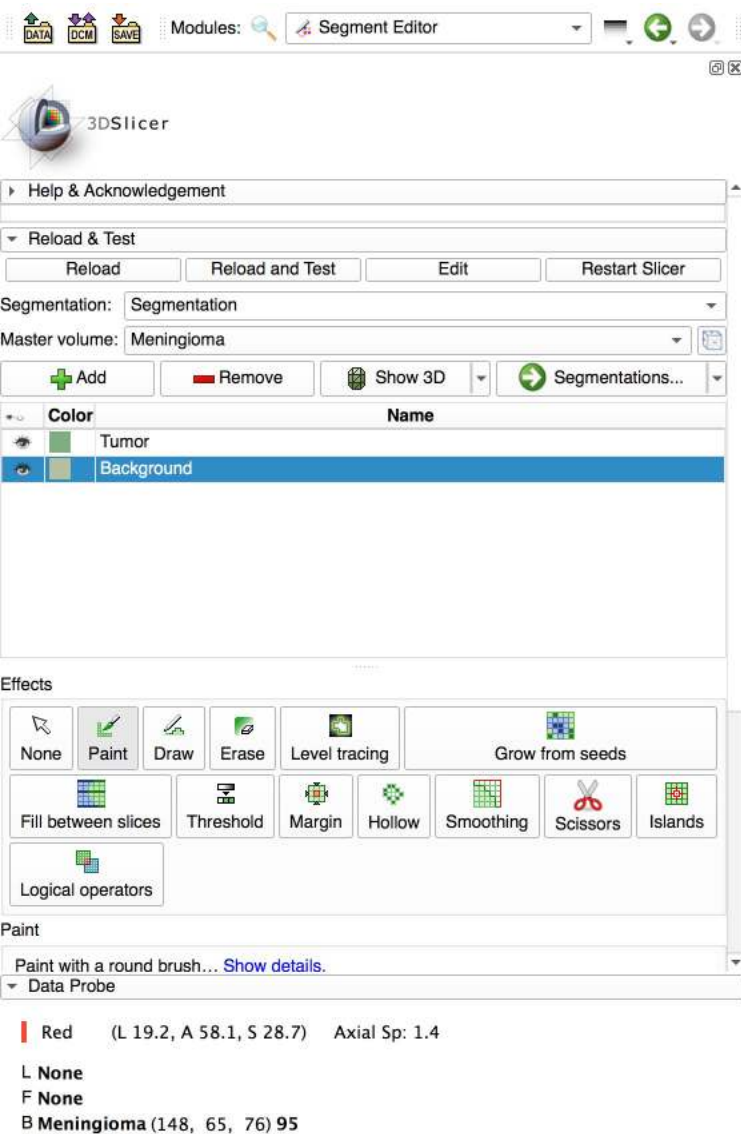
Módulo Segment Editor



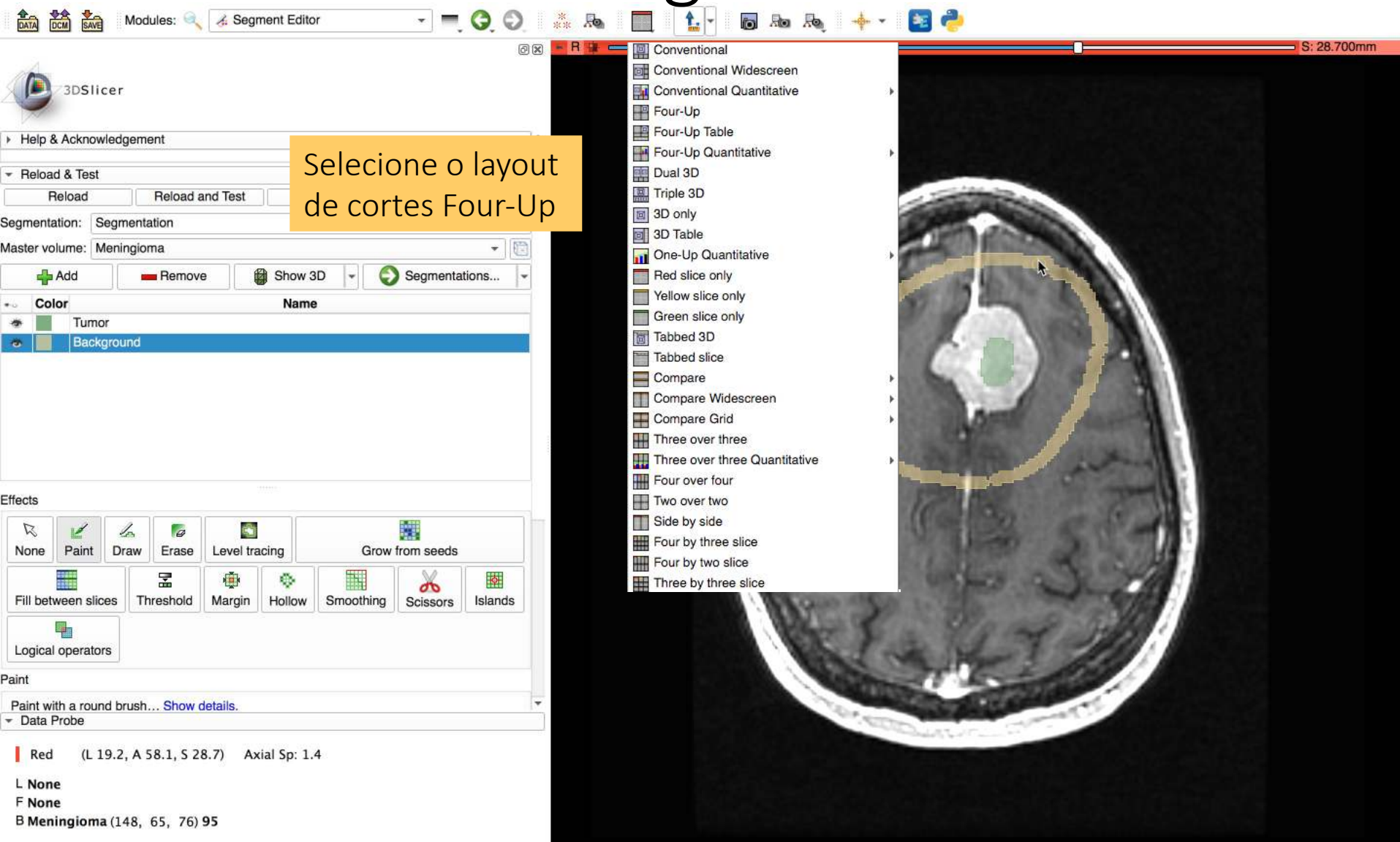
Módulo Segment Editor



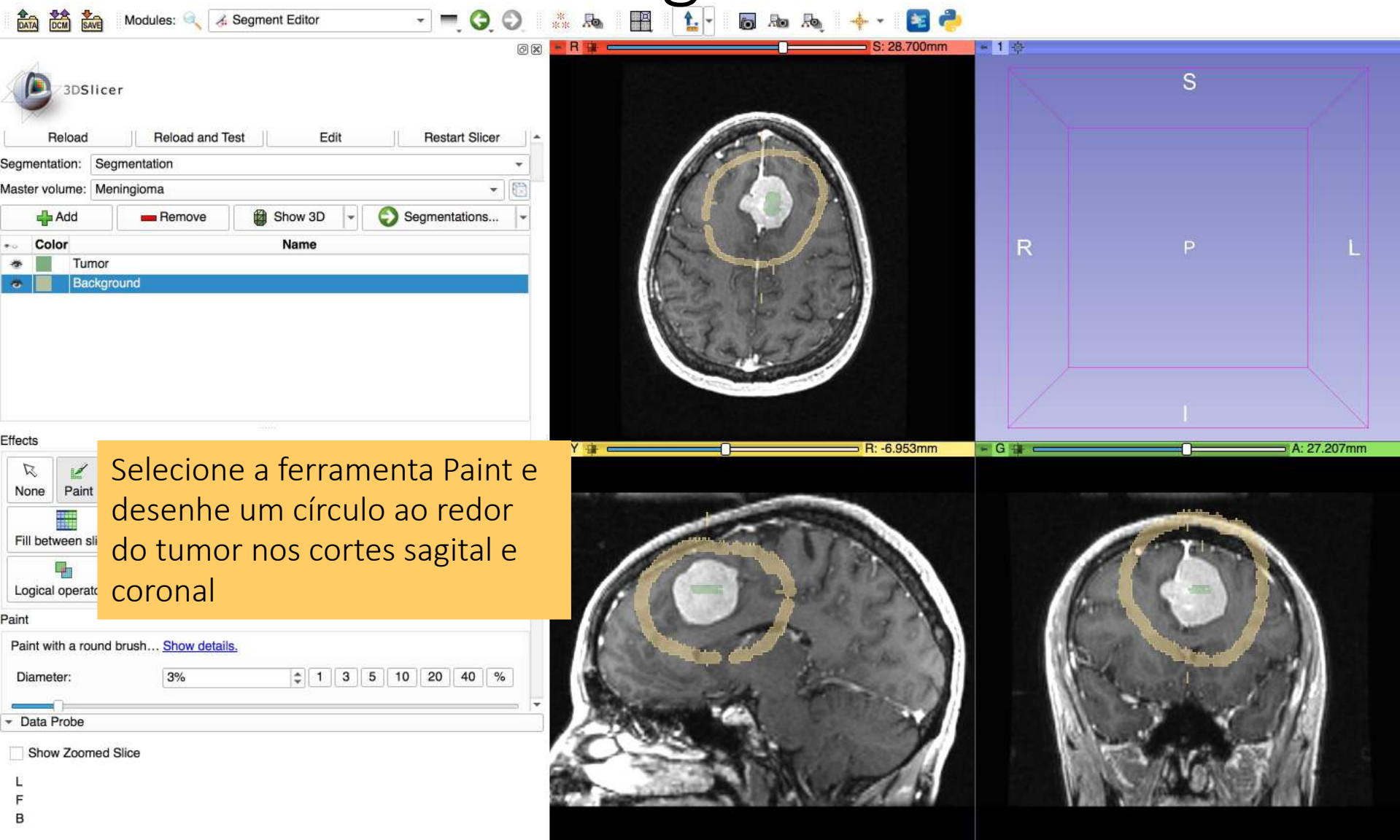
Módulo Segment Editor



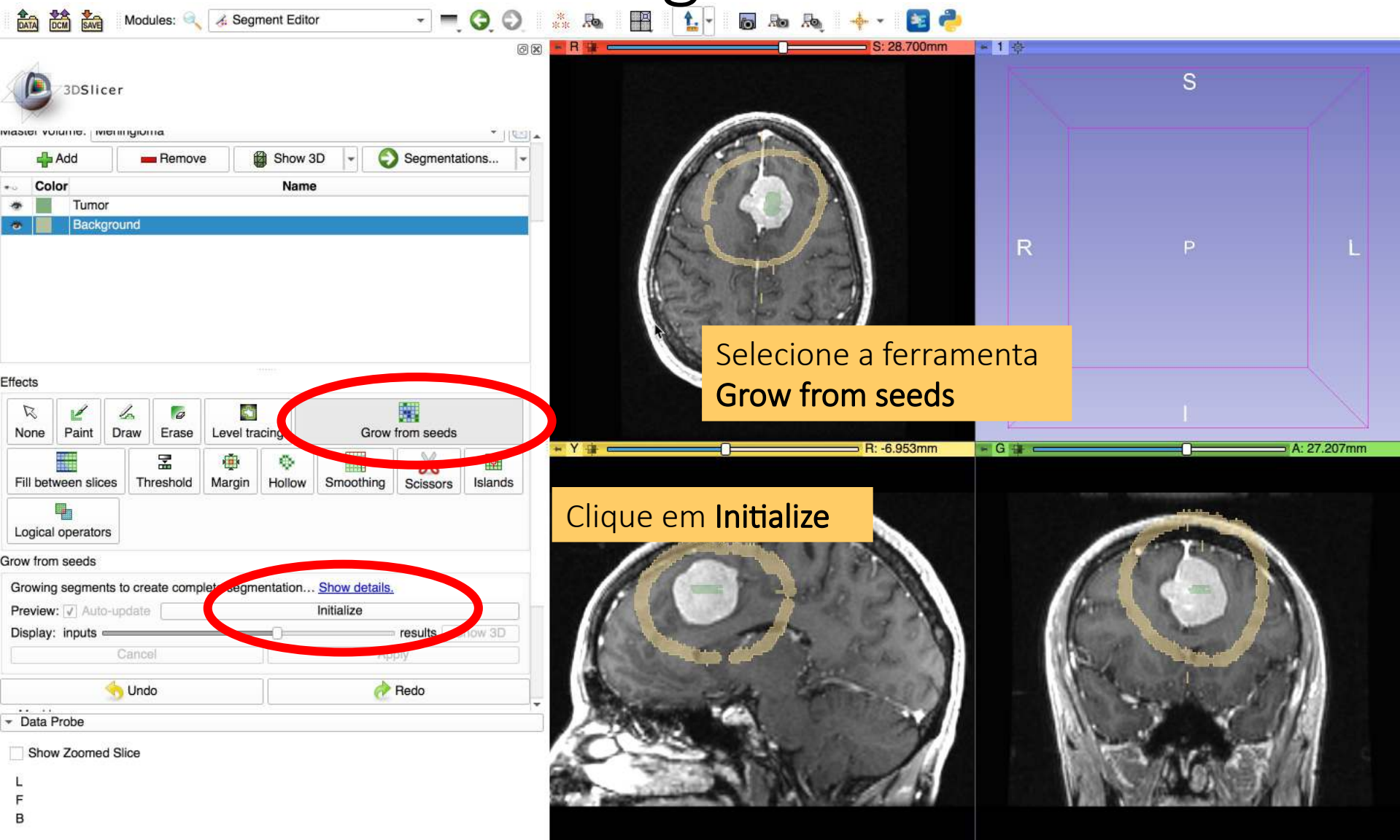
Módulo Segment Editor



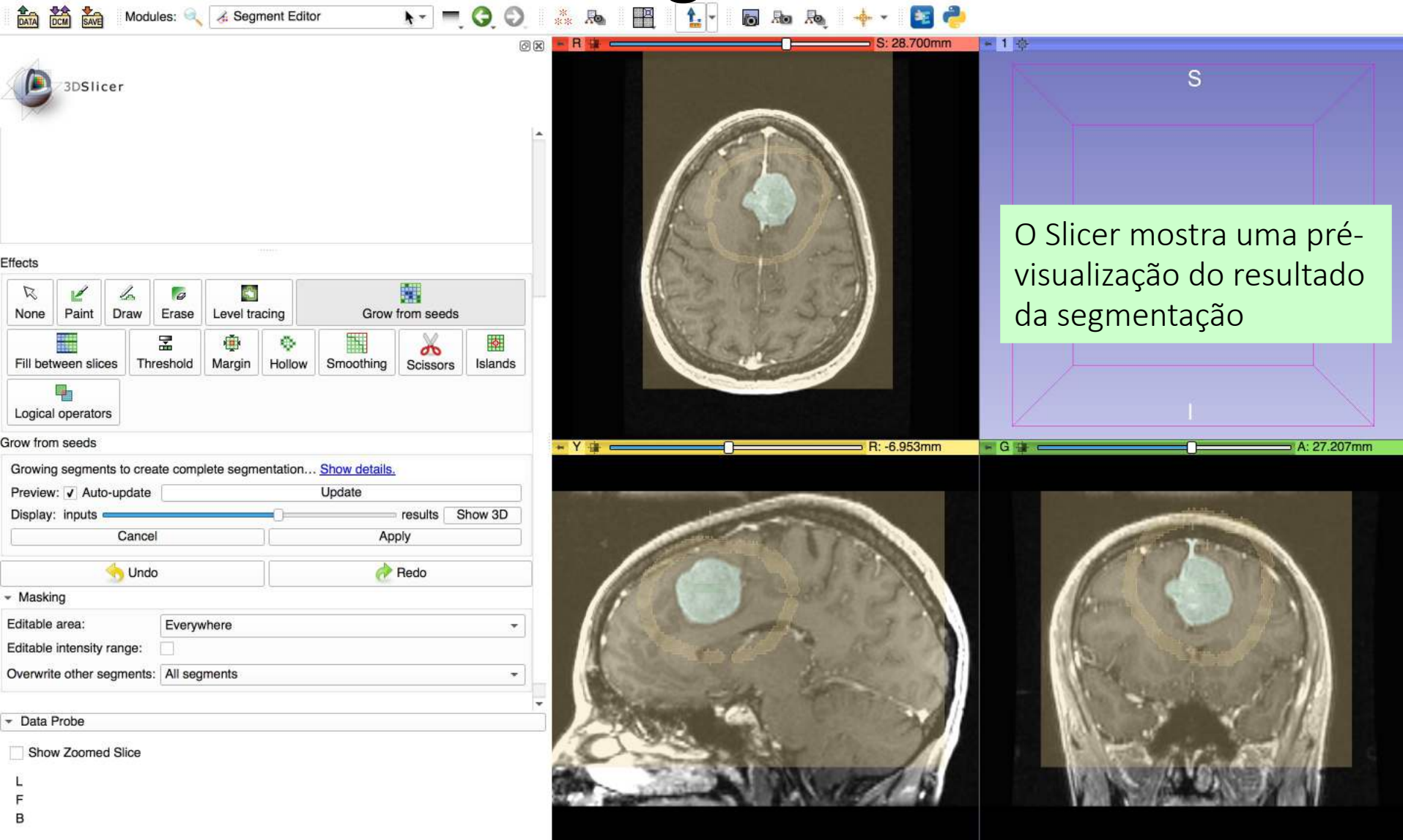
Módulo Segment Editor



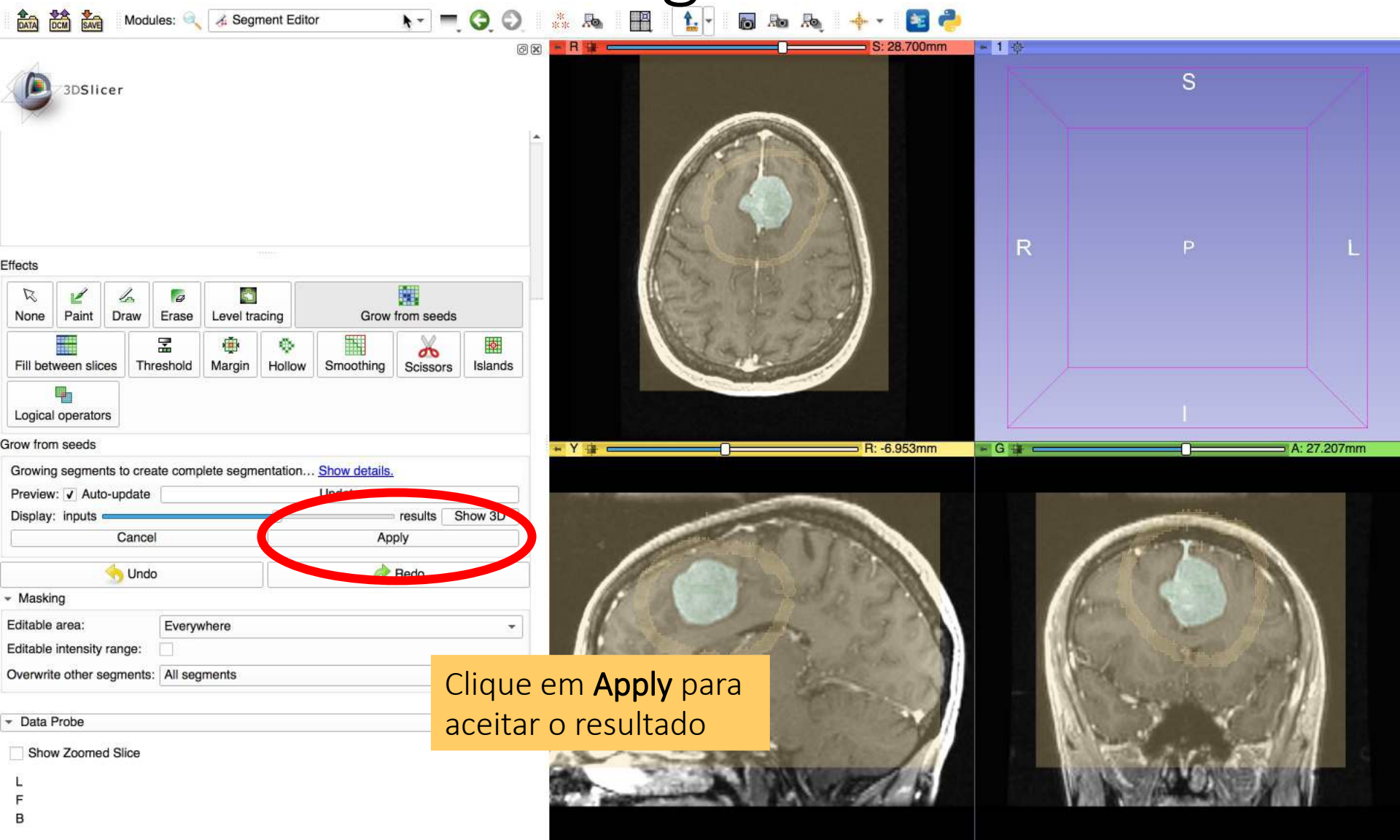
Módulo Segment Editor



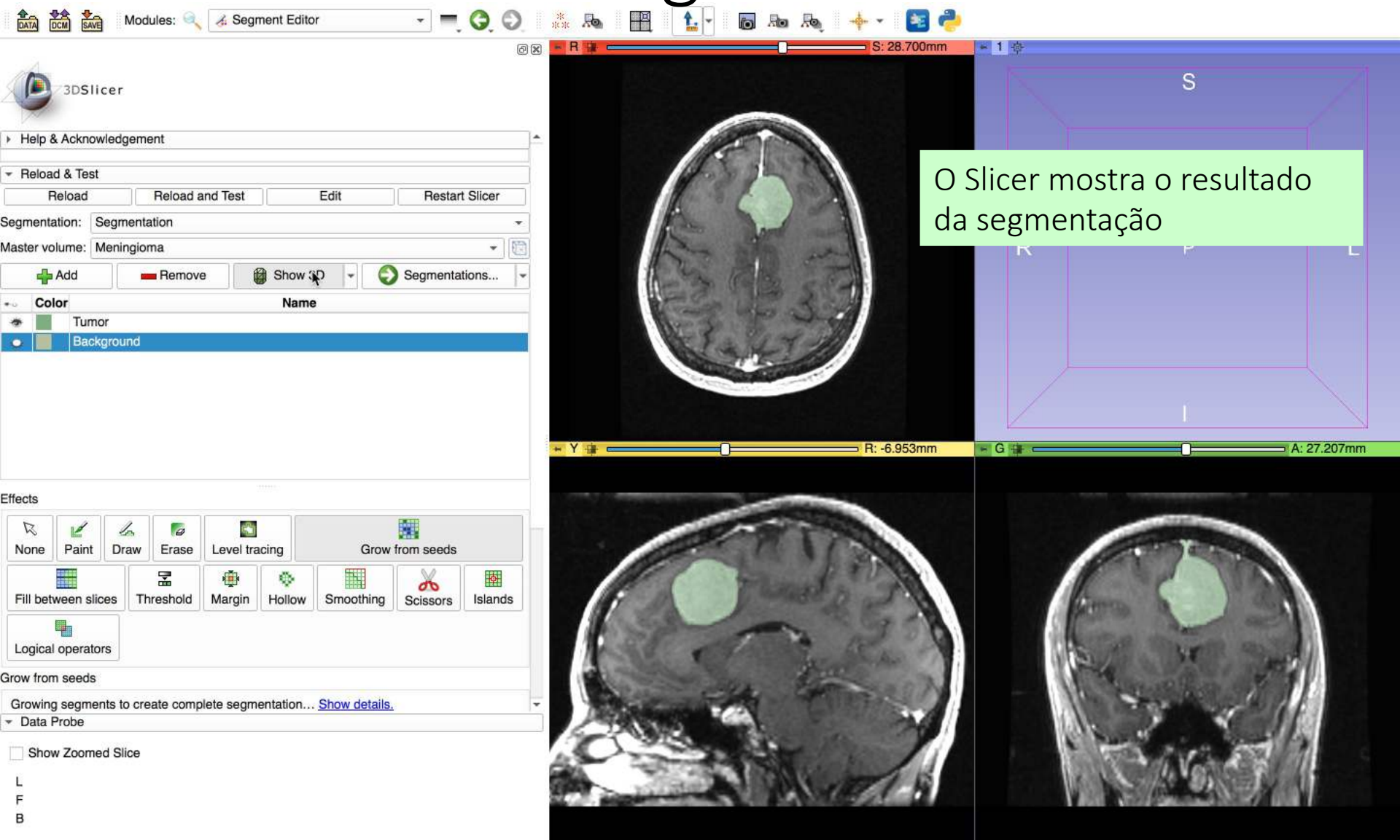
Módulo Segment Editor



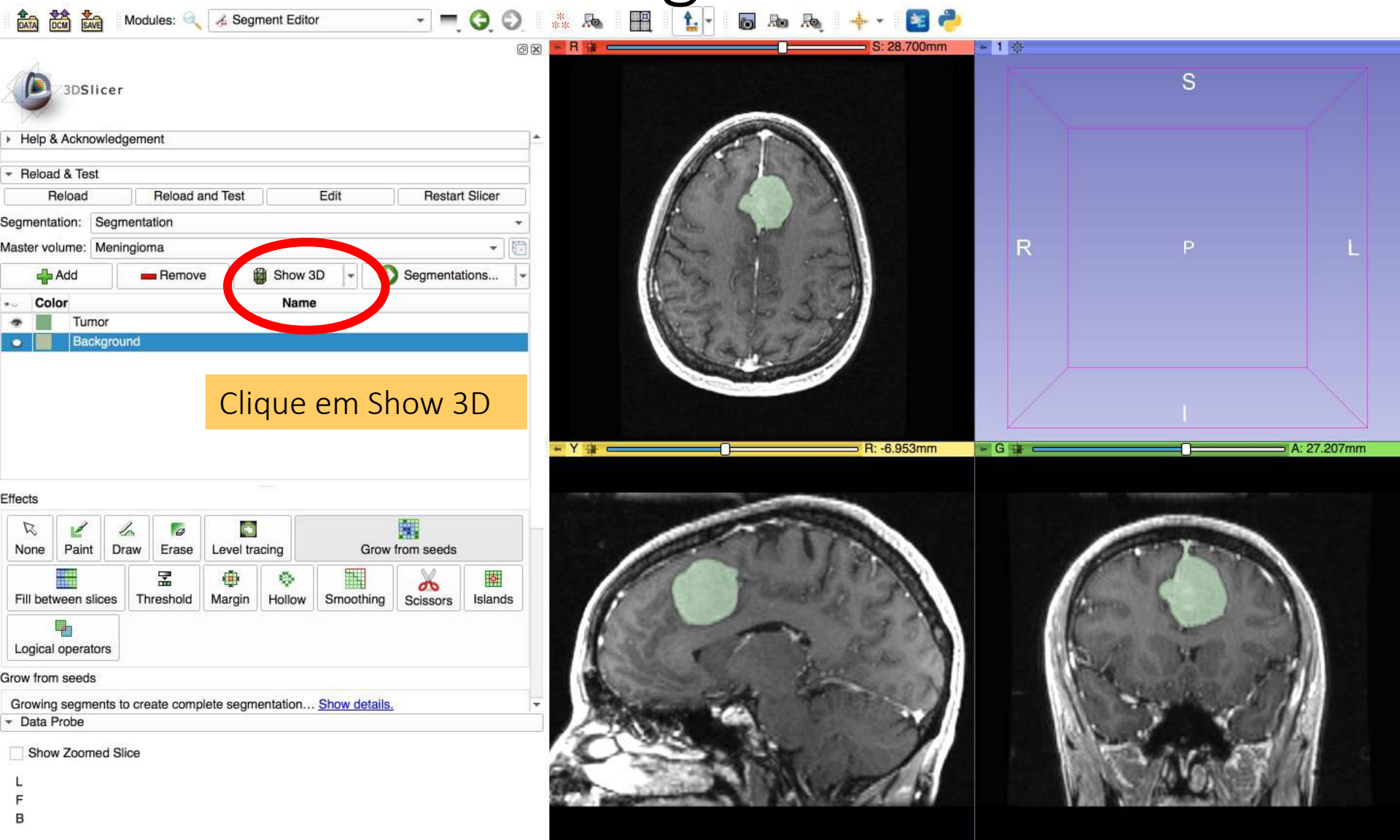
Módulo Segment Editor



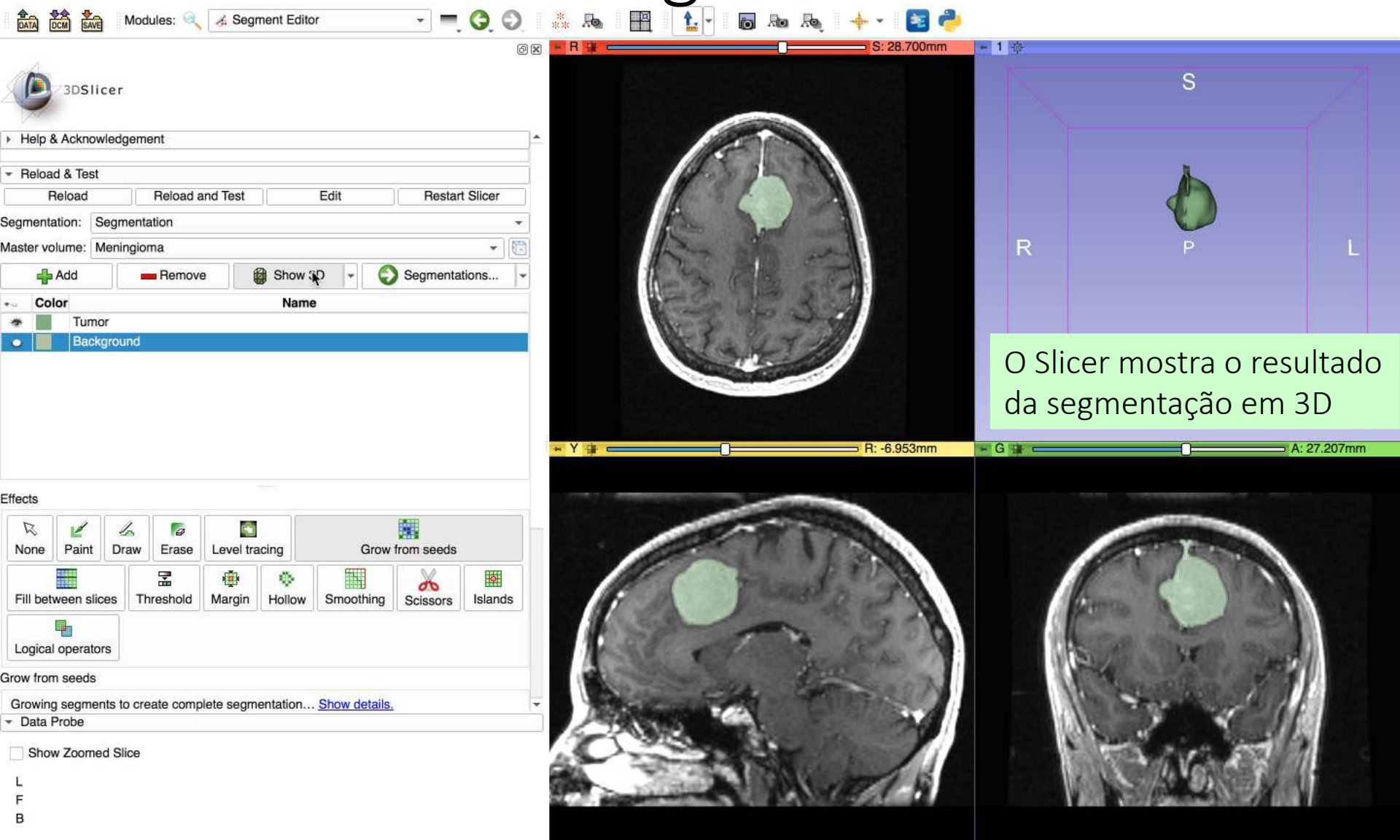
Módulo Segment Editor



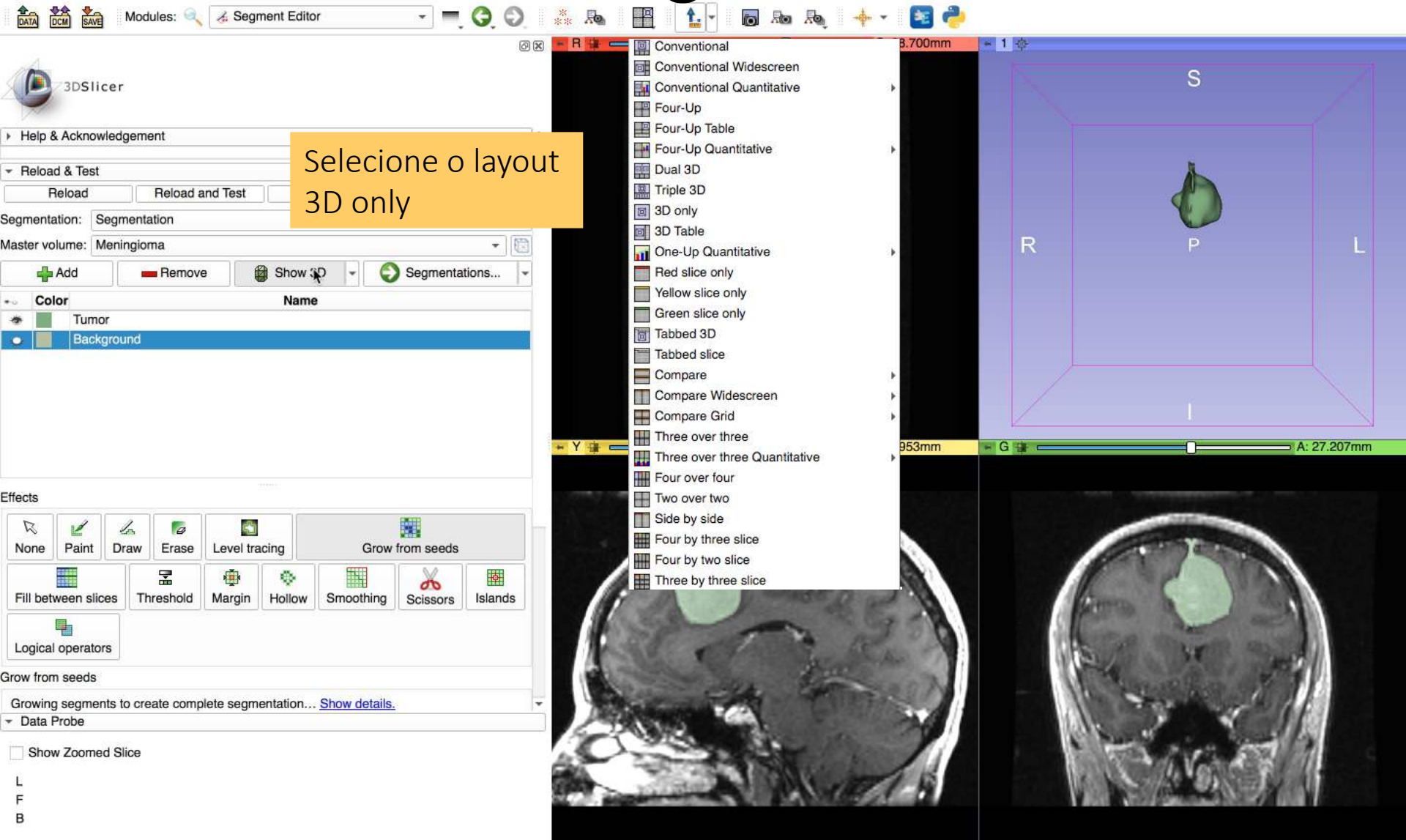
Módulo Segment Editor



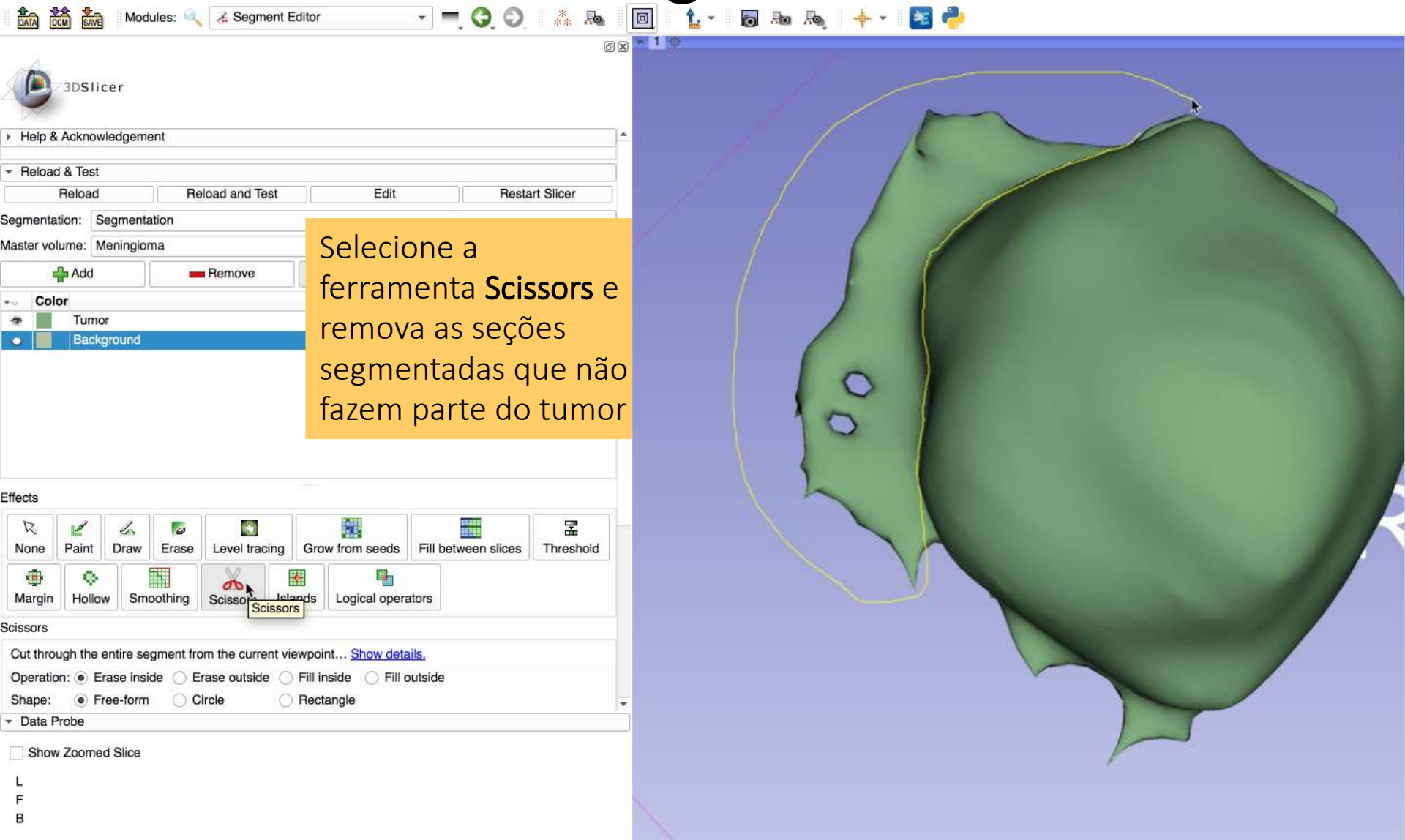
Módulo Segment Editor



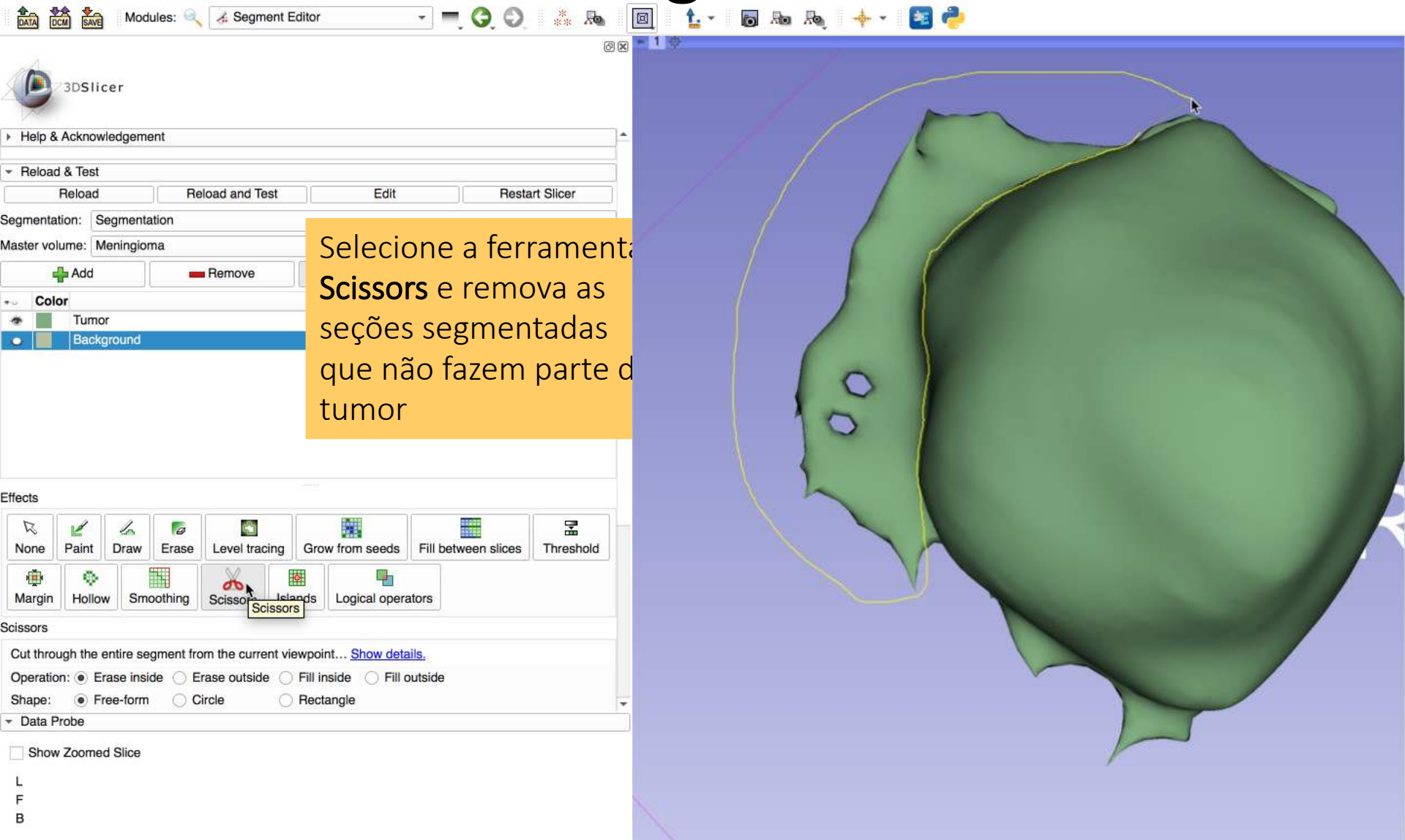
Módulo Segment Editor



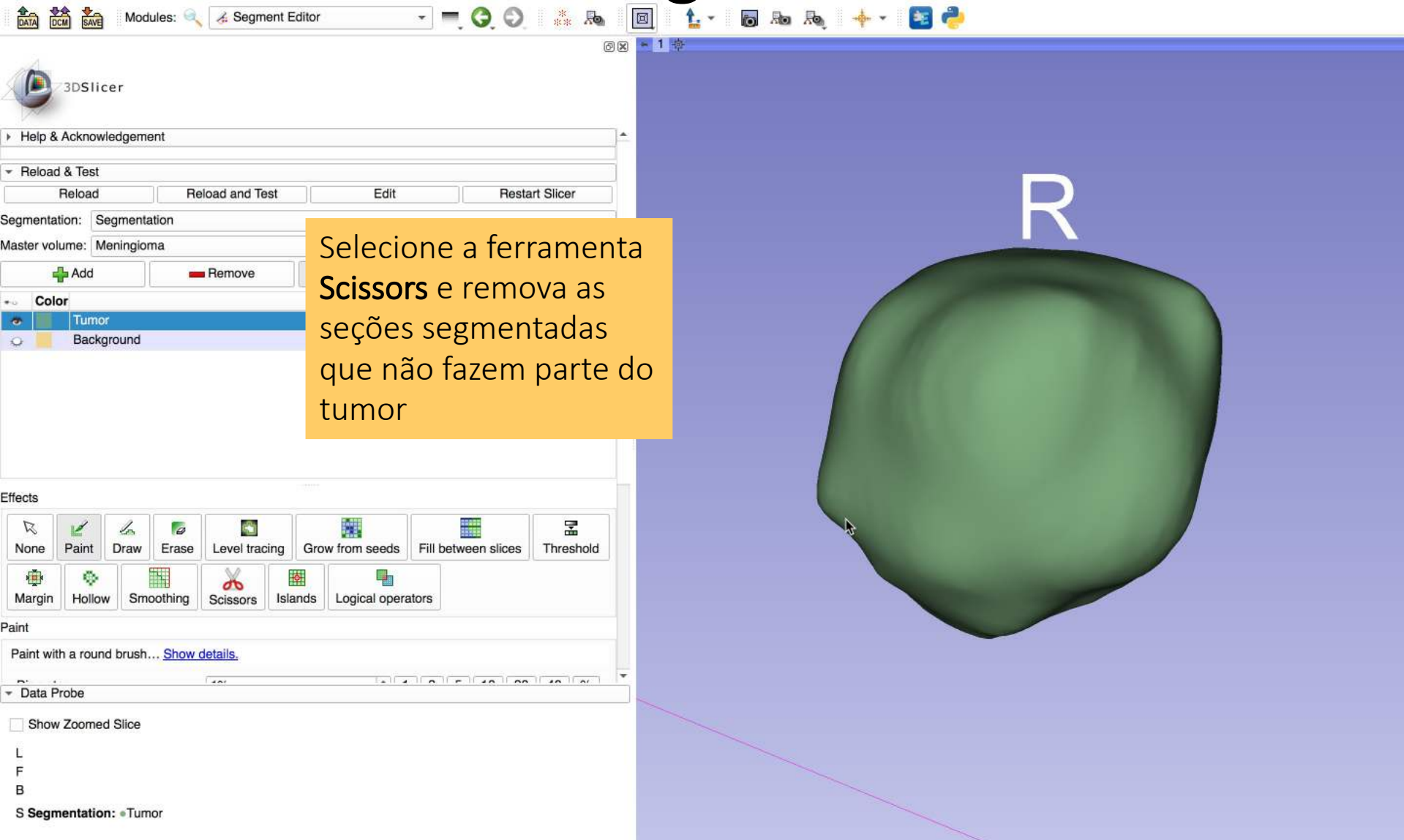
Módulo Segment Editor

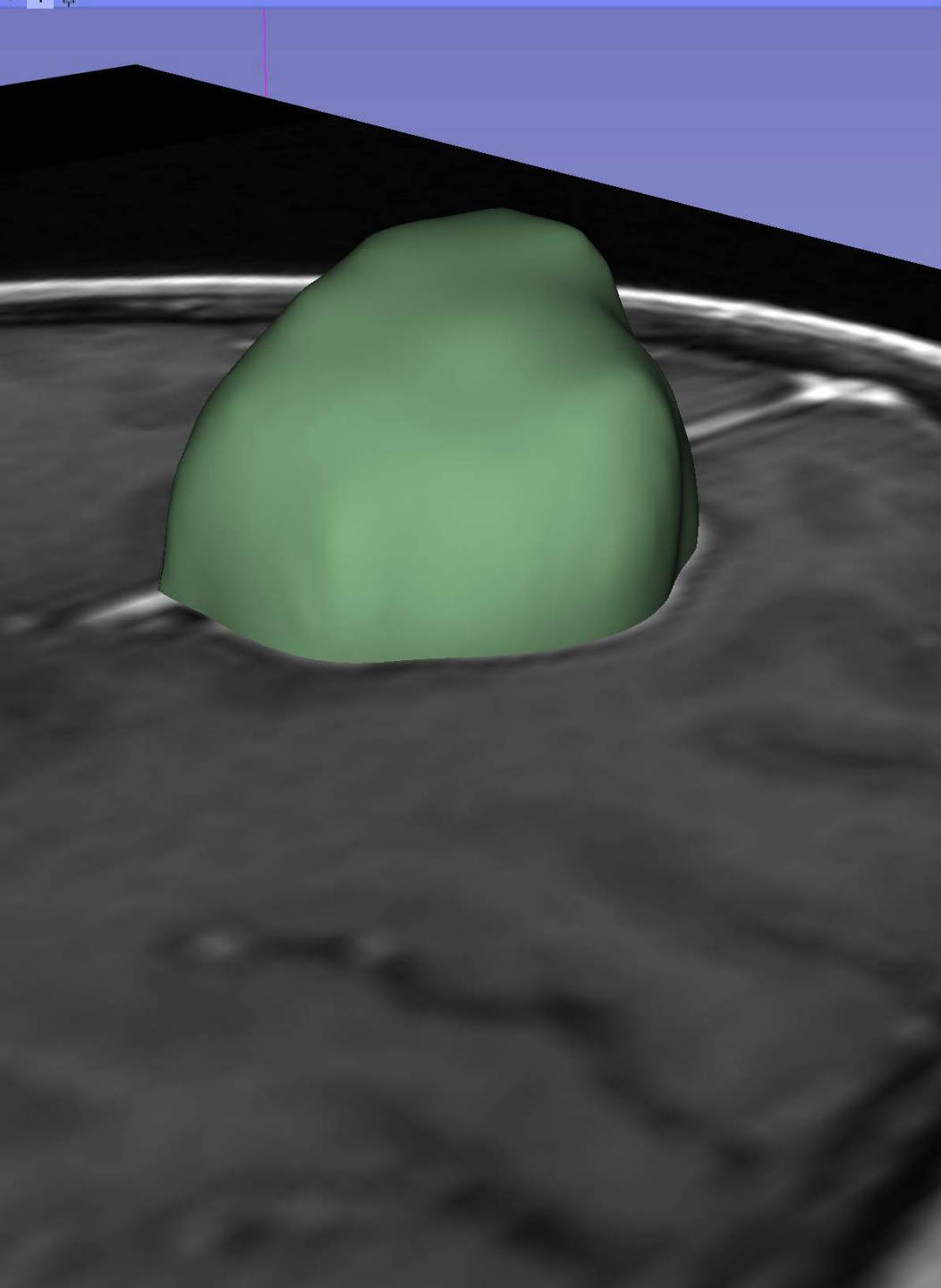


Módulo Segment Editor



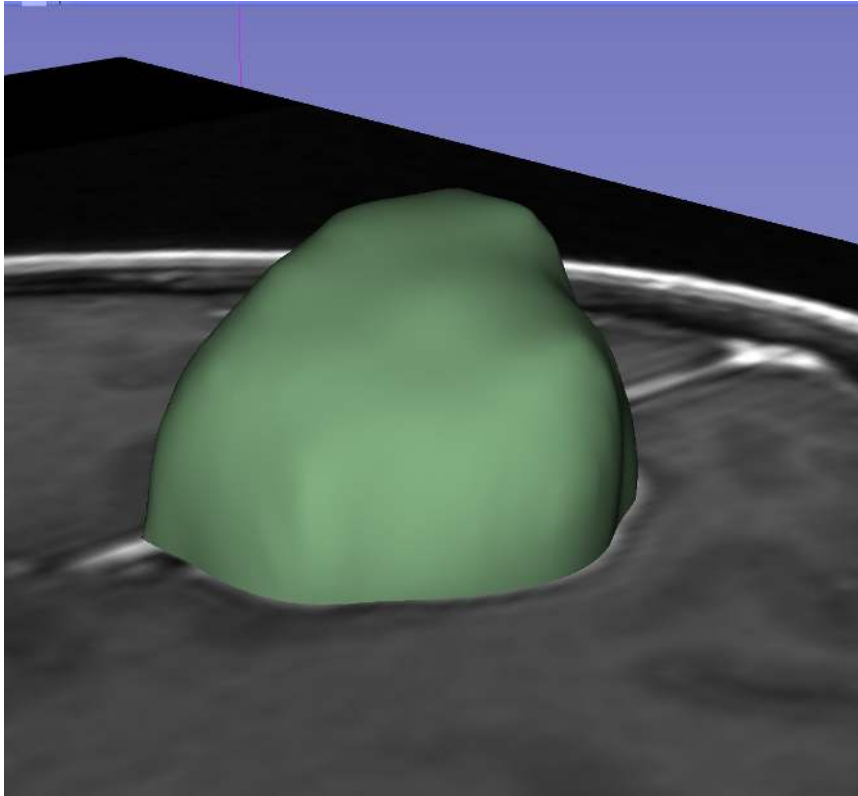
Módulo Segment Editor





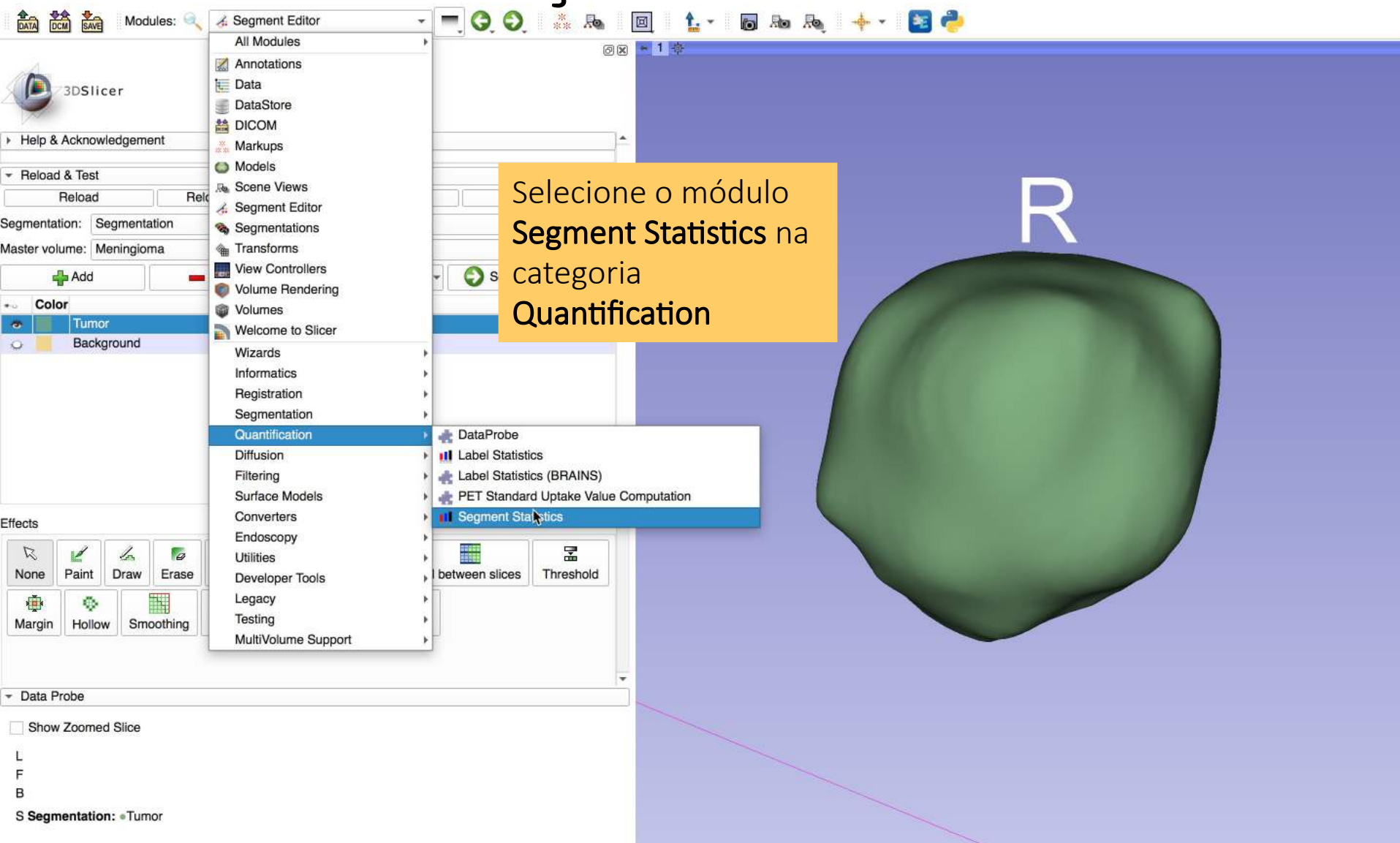
Parte 3: Cálculo do volume do tumor

Medições 3D

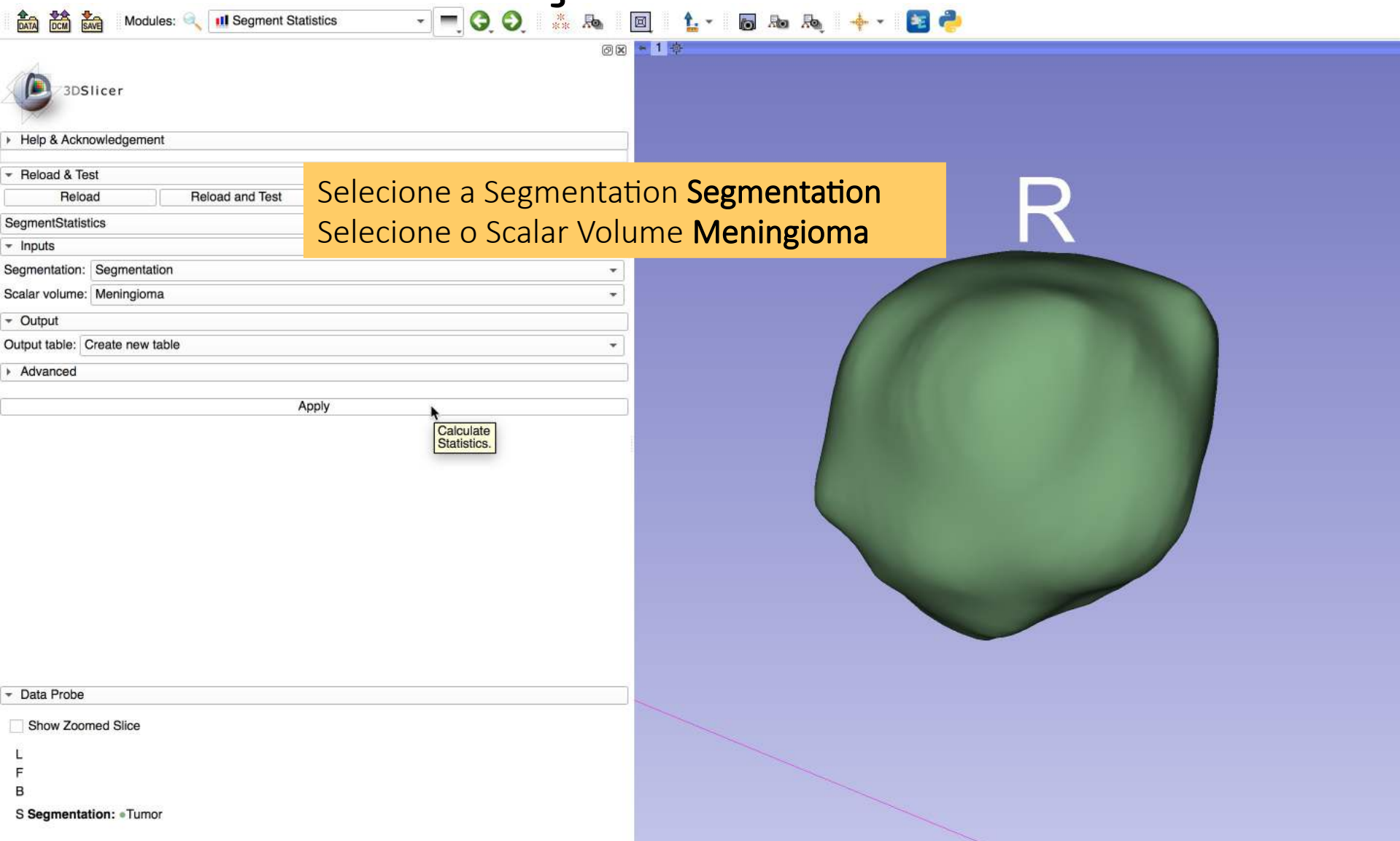


Esta seção mostra como calcular o volume do tumor segmentado usando o módulo Segment Statistics

Medições do tumor



Medições do tumor



Medições do tumor

The screenshot displays the 3D Slicer software interface. On the left, the 'Segment Statistics' module is active, showing a table with columns for 'Segmentation', 'Scalar volume', and 'Output'. The 'Segmentation' column is set to 'Segmentation', 'Scalar volume' to 'Meningioma', and 'Output table' to 'Table'. The 'Apply' button is circled in red. A red arrow points to the 'Conventional' layout option in the top toolbar. The main 3D view shows a green tumor model labeled 'R'. Below the 3D view, there are three 2D slice views (axial, sagittal, and coronal) showing the tumor in cross-section. The bottom status bar displays coordinates: Y, R: 0.469mm, G, A: 27.481mm.

Selezione o layout **Conventional** e clique em Apply

Medições do tumor

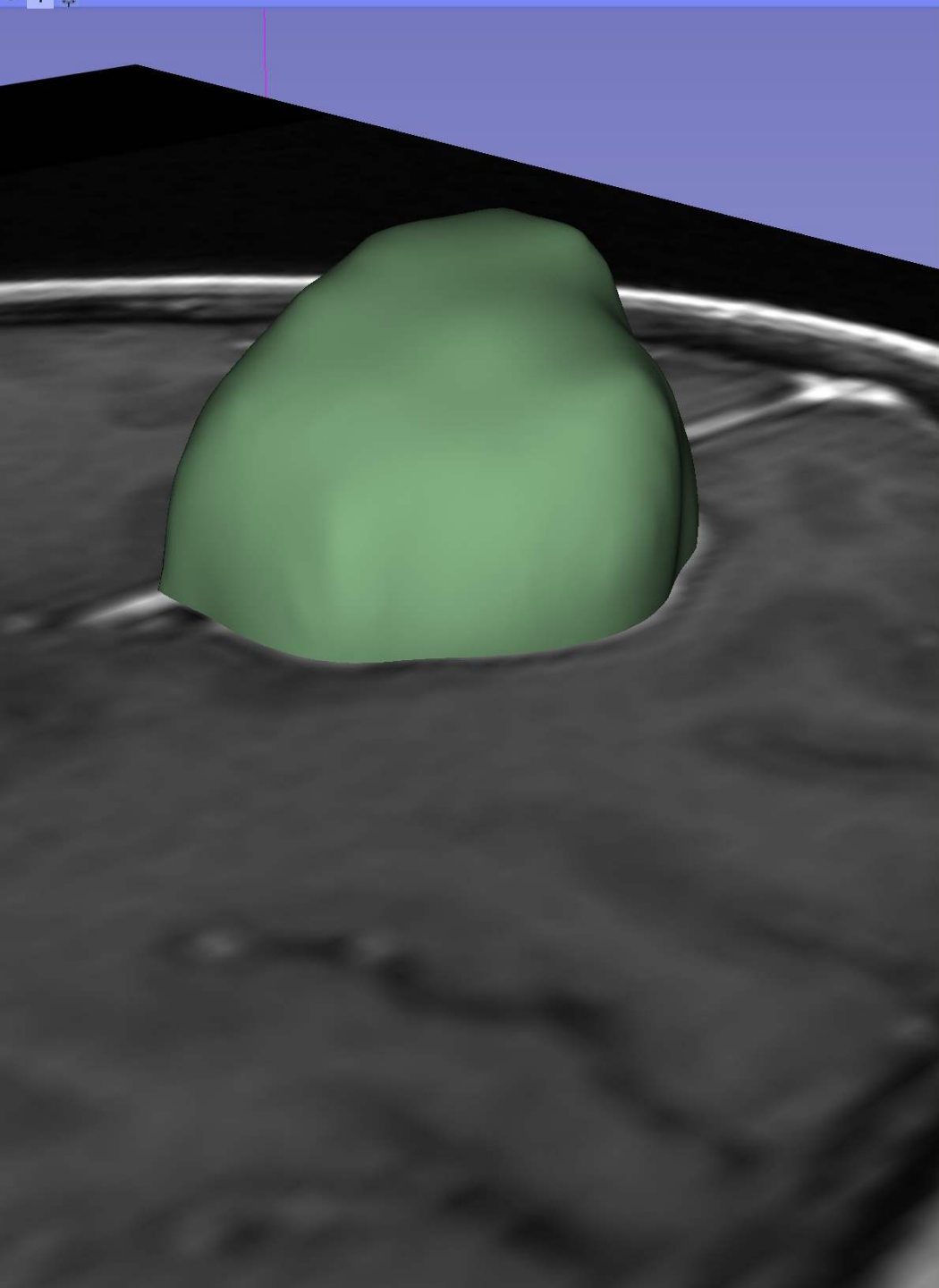
The screenshot displays the 3D Slicer interface with the Segment Statistics module active. The left sidebar shows the 'Segment Statistics' panel with the following settings:

- Segmentation: Segmentation
- Scalar volume: Meningioma
- Output table: Table

The main view shows three orthogonal slices (axial, sagittal, and coronal) of a brain MRI with a green tumor segmentation. A 3D model of the tumor is visible in the top right corner. The bottom right corner displays a table with the following data:

	A	B	C	D	E	F
1 Segment	Number of voxels [voxels] (1)	Volume [mm3] (1)	Volume [cm3] (1)	Number of voxels [voxels] (2)	Volume [mm3] (2)	
2 Tumor	13859	17053.1	17.0531	13859	17053.1	

The text box on the left indicates the volume calculated by Slicer: 17,053 cm³.



Parte 4: Cálculo das características quantitativas de imagem



Pacote PyRadiomics

- **Pyradiomics** é um pacote Python de código aberto que permite a extração automatizada de mais de 1.500 características quantitativas a partir de dados de imagens médicas.
- O pacote inclui ferramentas para pré-processamento e filtragem de imagens com base no SimpleITK
- As ferramentas de linha de comando do Pyradiomics permitem o processamento em lote

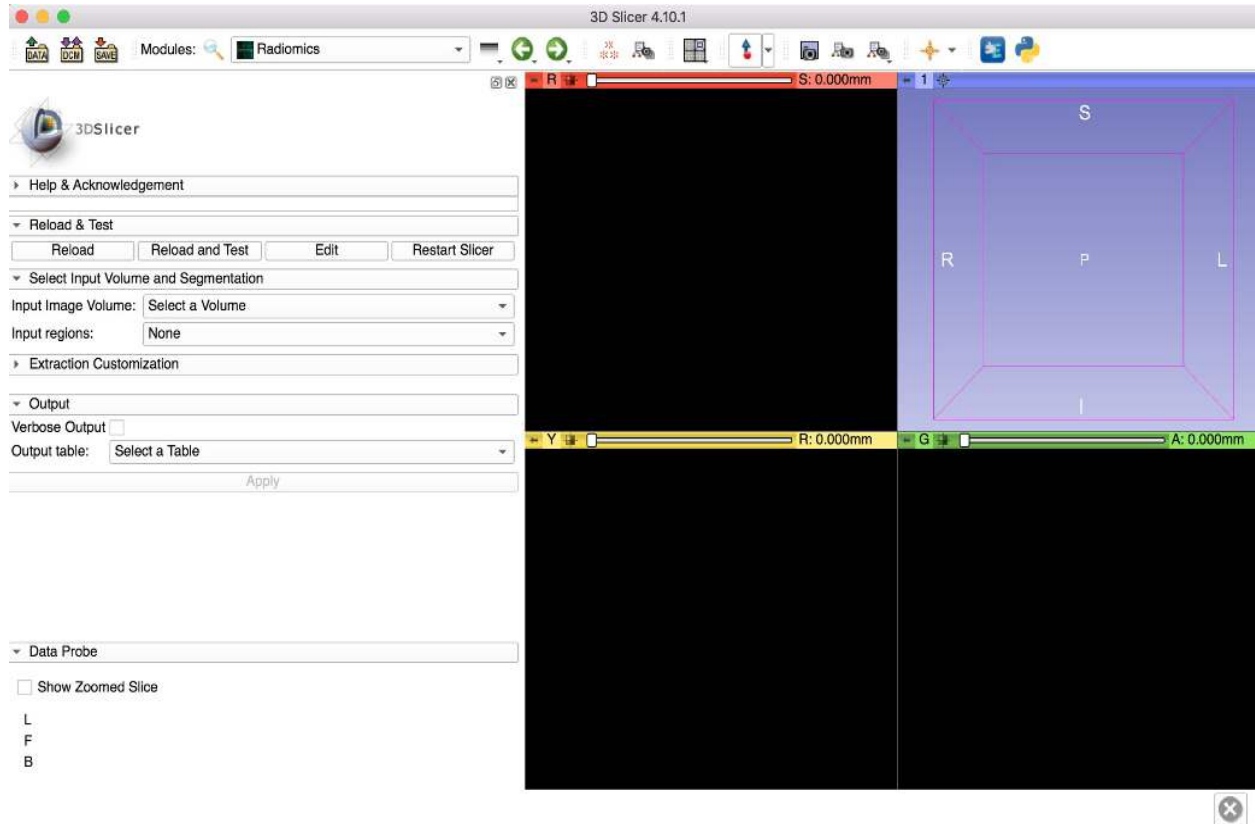
<https://pyradiomics.readthedocs.io>

<http://www.radiomics.io/>

van Griethuysen, J. J. M., Fedorov, A., Parmar, C., Hosny, A., Aucoin, N., Narayan, V., Beets-Tan, R. G. H., Fillon-Robin, J. C., Pieper, S., Aerts, H. J. W. L. (2017). Computational Radiomics System to Decode the Radiographic Phenotype. Cancer Research, 77(21), e104–e107.



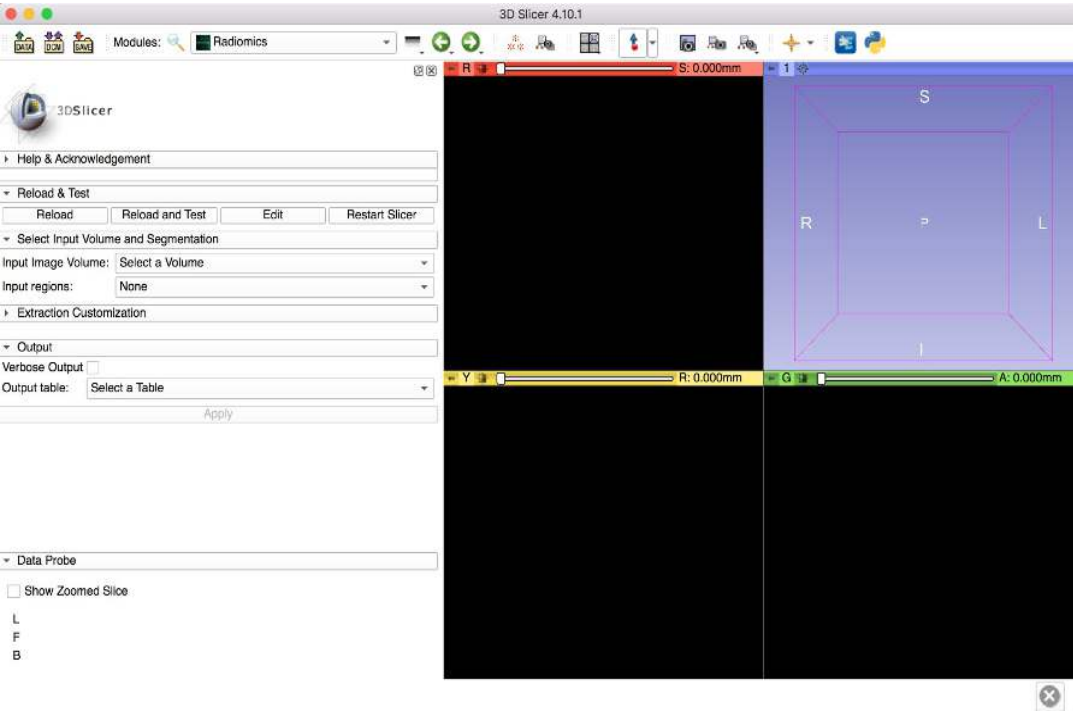
Slicer Radiomics



A extensão **Slicer Radiomics** fornece uma interface gráfica de usuário para a biblioteca pyradiomics

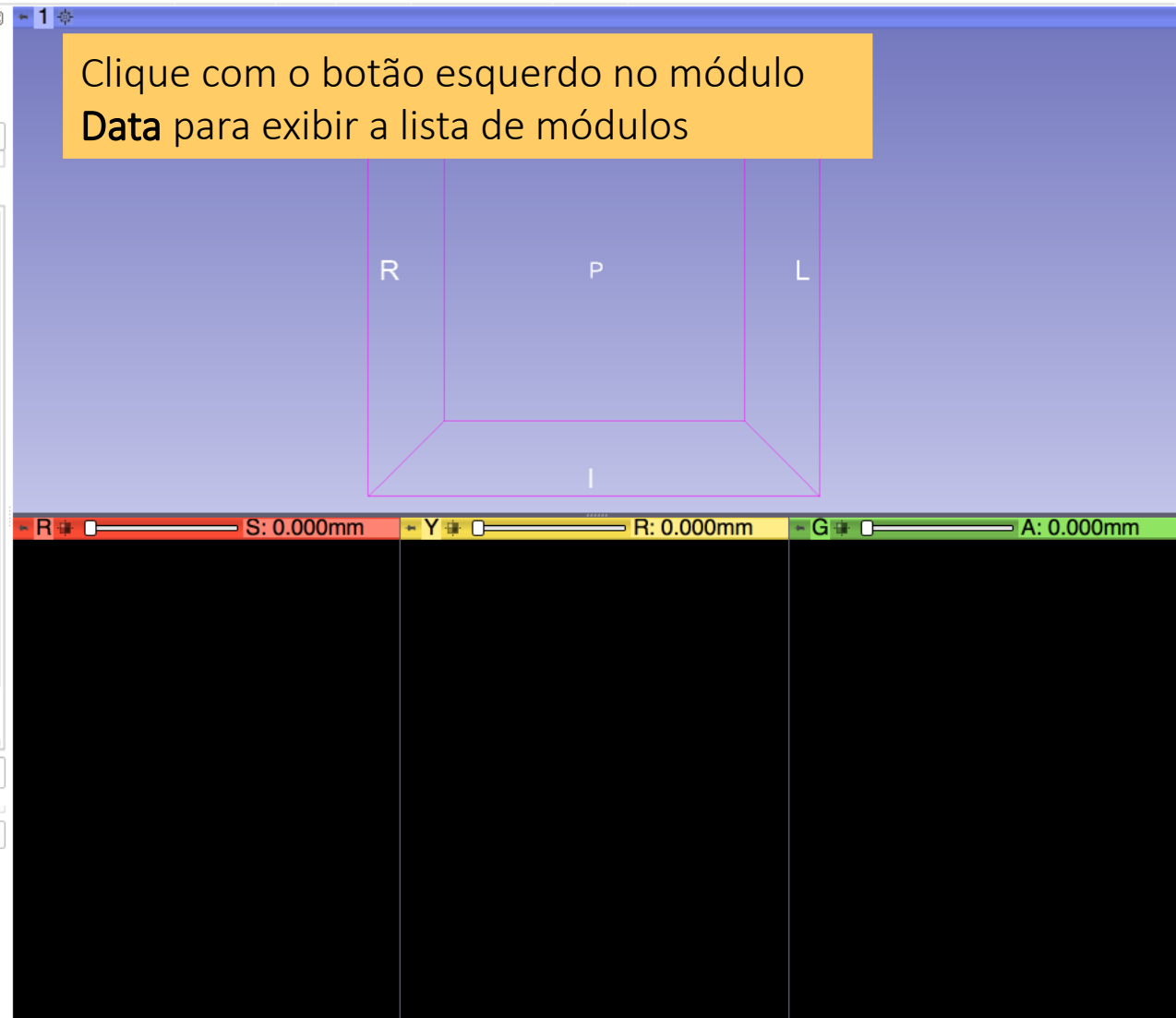
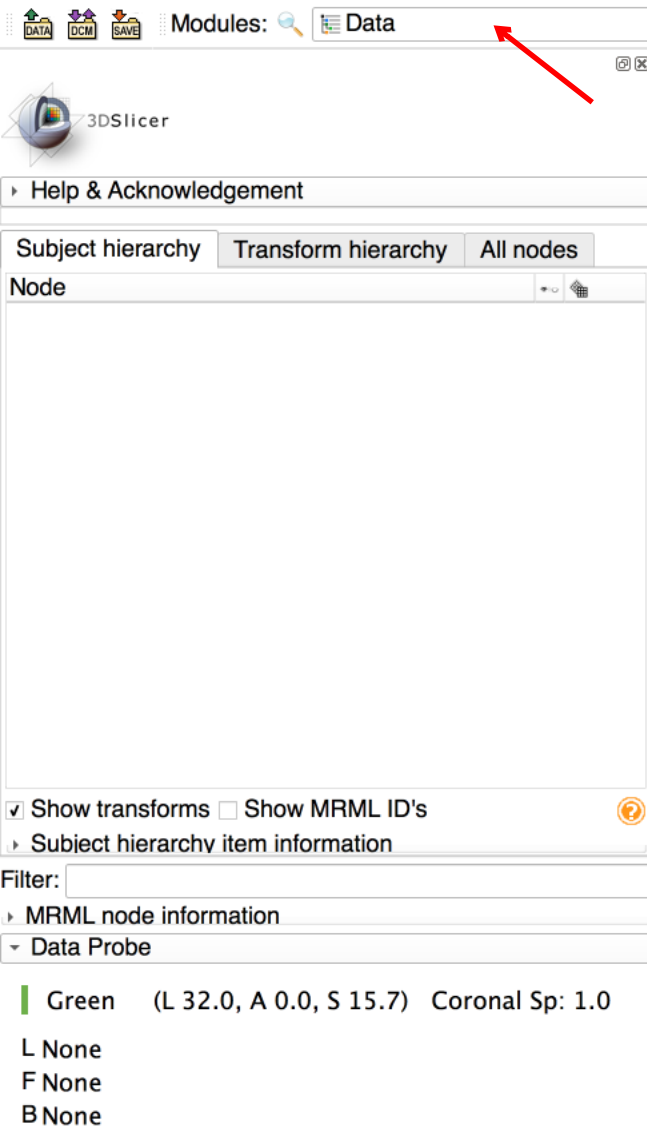


Slicer Radiomics

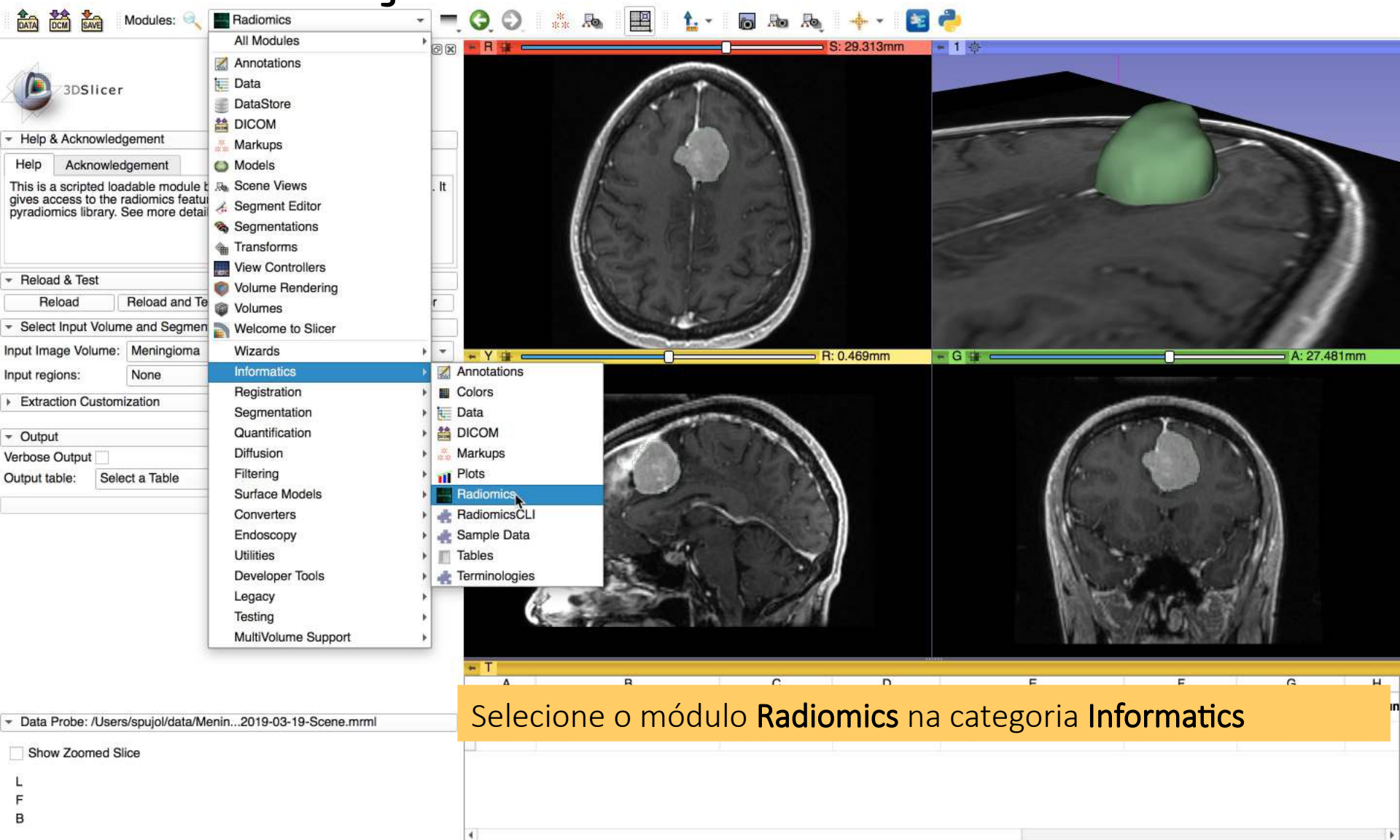


A extensão Slicer Radiomics permite a exploração interativa de características de imagem e a configuração dos parâmetros de extração

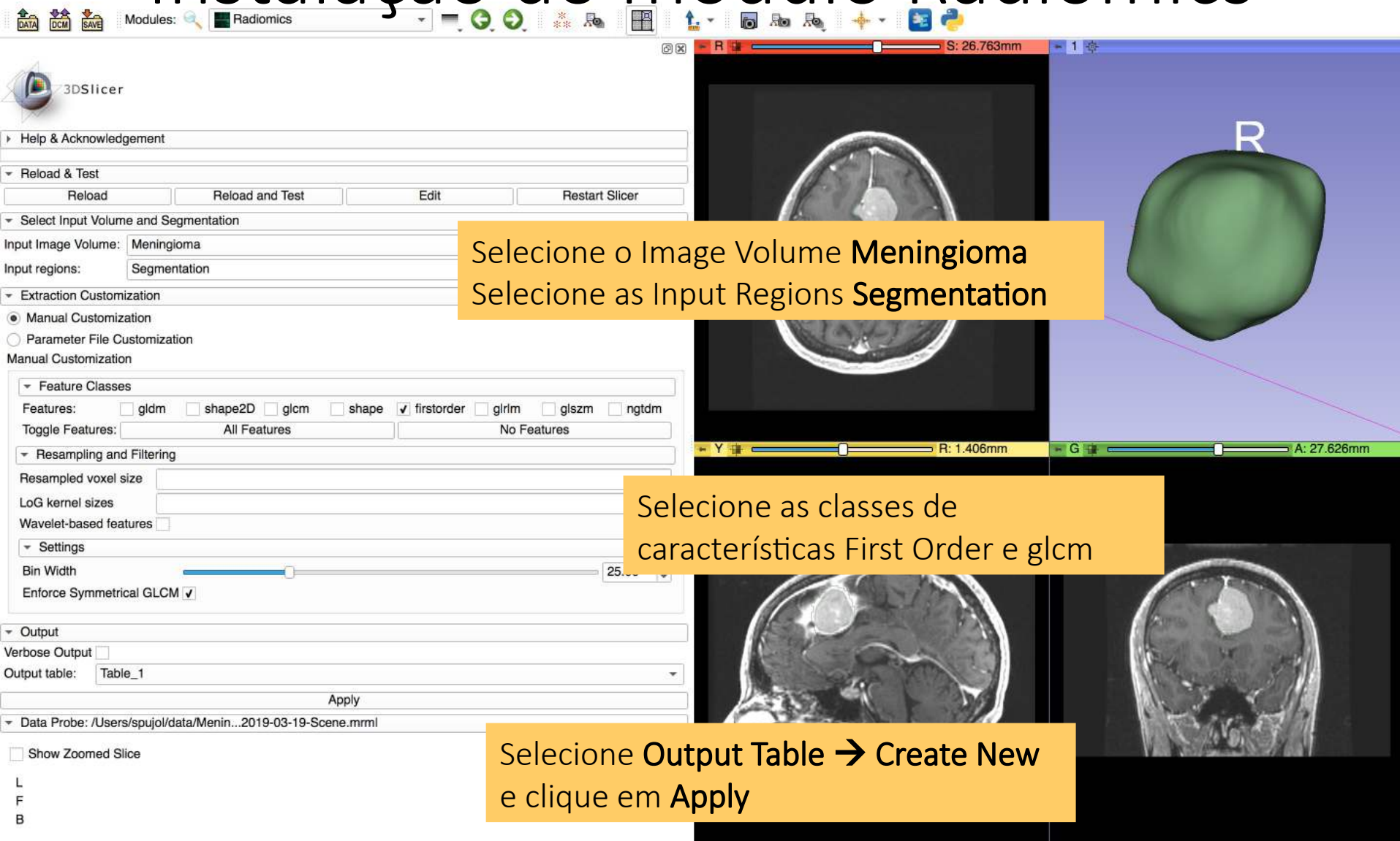
Módulo Radiomics



Instalação do módulo Radiomics



Instalação do módulo Radiomics



The screenshot shows the 3D Slicer Radiomics module interface. The left sidebar contains the 'Radiomics' module panel with sections for 'Help & Acknowledgement', 'Reload & Test', 'Select Input Volume and Segmentation', 'Extraction Customization', 'Feature Classes', 'Resampling and Filtering', 'Settings', 'Output', and 'Data Probe'. The main window displays three orthogonal views of a brain MRI scan: axial, sagittal, and coronal. A green 3D model of the segmented tumor is visible in the top right corner. The 'Radiomics' module panel is currently set to 'Manual Customization' and 'Feature Classes' are selected. The 'Feature Classes' section shows 'Features' with 'firstorder' and 'glcm' checked, and 'Toggle Features' set to 'All Features'. The 'Resampling and Filtering' section shows 'Resampled voxel size' set to 25.0. The 'Output' section shows 'Output table' set to 'Table_1'. The 'Data Probe' section shows the file path '/Users/spujol/data/Menin...2019-03-19-Scene.mrml'.

Selezione o Image Volume **Meningioma**
Selezione as Input Regions **Segmentation**

Selezione as classes de características First Order e glcm

Selezione Output Table → Create New e clique em **Apply**

Cálculo das características

The screenshot displays the 3D Slicer Radiomics module interface. On the left, the 'Radiomics' panel is visible, showing the 'Manual Customization' tab. The 'Input Image Volume' is set to 'Meningioma' and the 'Input regions' to 'Segmentation'. The 'Feature Classes' section includes checkboxes for 'gldm', 'shape2D', 'gldm', 'shape', 'firstorder' (checked), 'gldm', and 'gldm'. The 'Resampling and Filtering' section shows 'Resampled voxel size' as 25.00. The 'Output' section shows 'Verbose Output' as 'Table_1'. The 'Data Probe' shows the file path: '/Users/spujol/data/Menin...2019-03-19-Scene.mrml'. The main window shows a brain MRI with a green segmentation of a meningioma. The 'Radiomics' table at the bottom right lists the extracted features and their values.

O Slicer exibe o resultado do cálculo das características

	A	B	C	D	E
	Image type	Feature Class	Feature Name	Segmentation_segment_Tu...	Segmentation_segme
1	diagnostics	Versions	PyRadiomics	2.1.2.post58+g6d385a1	2.1.2.post58+g6d385a1
2	diagnostics	Versions	Numpy	1.13.1	1.13.1
3	diagnostics	Versions	SimpleITK	1.1.0	1.1.0
4	diagnostics	Versions	PyWavelet	1.0.0	1.0.0
5	diagnostics	Versions	Python	2.7.13	2.7.13
6	diagnostics	Configuration	Settings	{'distances': [1], 'additionalInfo': True, 'force2D': False, 'interpol...	{'distances': [1], 'additio...
7	diagnostics	Configuration	EnabledImageTypes	{'u'Original': {}}	{'u'Original': {}}
8	diagnostics	Image-original	Hash	d96124aaa2dcfd45fa2b4b80f6...	d96124aaa2dcfd45fa2b4b80f6...

Cálculo das características

O Slicer exibe os valores das 18 características da classe **FirstOrder** inclui 18 características

34	original	firstorder	InterquartileRange	20.0
35	original	firstorder	Skewness	1.99316623652
	original	firstorder	Uniformity	0.358800750225
	original	firstorder	Median	177.0
	original	firstorder	Energy	453513881.0
	original	firstorder	RobustMeanAbsoluteDeviation	9.02739158026
40	original	firstorder	MeanAbsoluteDeviation	15.6561086211
41	original	firstorder	TotalEnergy	558034658.262
42	original	firstorder	Maximum	370.0
43	original	firstorder	RootMeanSquared	180.8961587
44	original	firstorder	90Percentile	202.0
45	original	firstorder	Minimum	36.0
46	original	firstorder	Entropy	1.83048891173
47	original	firstorder	Range	334.0
48	original	firstorder	Variance	575.802209976
49	original	firstorder	10Percentile	156.0
50	original	firstorder	Kurtosis	12.4315749384
51	original	firstorder	Mean	179.297568367

Conclusão

- Este tutorial oferece uma introdução básica à fenotipagem de imagens usando a extensão Slicer Radiomics.
- A extensão permite o cálculo das classes de características implementadas no pyradiomics
- A descrição de cada classe está disponível em <https://pyradiomics.readthedocs.io/>

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